Conditional Cramér–Rao Bound Analysis of Medium-Frequency Radar Horizontal Wind Retrieval and Comparison With Full Correlation Analysis

Yang Zhang, Li-Ming Wang, *Member, IEEE*, and Na Li

***Abstract*—Full correlation analysis (FCA) is widely used for horizontal wind retrieval with medium-frequency (MF) radar. This study derives an analytical Fisher-information matrix (FIM) and Cramér–Rao bound (CRB) from the adopted full-echo likelihood under a fixed and known scattering realization. Under the corresponding regularity and unbiasedness conditions, the CRB provides a conditional covariance lower bound that complements existing variance and precision analyses of specified FCA/spaced-antenna estimators. A conditional full-signal maximum-likelihood estimator provides numerical evidence of high-signal-to-noise ratio (SNR) tightness. Conventional FCA is evaluated relative to this benchmark, although the resulting gap cannot be attributed solely to processing loss because the benchmark assumes additional scattering-realization knowledge. The local full-signal correction analysis quantifies the reference-wind accuracy required to remain near the conditional bound. In the investigated configuration, the mismatch allowed by a 10%-near-CRB criterion decreases from approximately 0.424 m/s at 0 dB to 0.105 m/s at 30 dB. An FCA-initialized correction improves the retrieval error but remains above this conditional benchmark. Measured echoes provide qualitative checks of selected observable characteristics and illustrate externally assisted correction; they do not independently validate the conditional CRB. The results establish a model-specific information benchmark and clarify the reference-accuracy requirements for approaching it under prescribed scattering knowledge.**

***Index Terms*—Best linear unbiased estimator (BLUE), Cramér–Rao bound (CRB), error analysis, full correlation analysis (FCA), maximum-likelihood estimator (MLE), medium-frequency (MF) radar.**

I. Introduction

ATMOSPHERIC dynamics aims to elucidate the kinetic mechanisms of atmospheric motion and the physical laws governing climate evolution. As a result, accurate wind-field measurement in the mesosphere and lower thermosphere (MLT) is essential for understanding upper atmospheric circulation, tides, planetary waves, and gravity-wave transport [1], [2], [3], [4], [5], [6]. Among the currently available remote-sensing techniques, routine MLT observations are commonly obtained from meteor and medium-frequency (MF) radars, optical systems such as sodium temperature/wind lidars and nightglow imagers, and satellite limb/emission instruments [7], [8], [9], [10]. Meteor radars now support network-scale wind retrieval, sodium lidars provide high-precision local wind and temperature profiles in the mesopause region, nightglow imagers resolve wave structures over wide horizontal fields of view, and recent satellite optical missions extend these observations to regional-to-global temperature and wave diagnostics [1], [8], [9], [10]. Within this broader observing landscape, meteor radar, and MF radar are widely used tools for routine wind retrieval in this height region [7], [11], [12]. In particular, MF radar is attractive because of its low operational cost, continuous all-weather capability, and high temporal resolution [13], [14]. These characteristics make MF radar an important instrument for long-term MLT dynamical studies. However, one critical issue remains. The wind retrieval error is not negligible.

The need for statistical error characterization is particularly clear in MLT dynamics research. Wind variance and related second-order statistics are often used to characterize gravity-wave activity. Their seasonal and height-dependent behavior is commonly interpreted in terms of background-wind filtering and momentum transport in the middle atmosphere [15], [16]. If wind retrieval error contributes significantly to the observed variance, the inferred dynamical variability can be biased. The associated spectra can also be distorted, as can the diagnosed modulation of tides or planetary waves. Published MF radar studies already show this effect. Gibbins and Kavanagh showed that, for Antarctic MF radars using FCA, the width of the wind distribution and the occurrence of outliers are strongly linked to anisotropy and changing ionization conditions [17]. They further showed that the measured variance need not be controlled primarily by gravity waves. Kudeki et al. [18] showed that wind-retrieval biases under inhomogeneous flow can contaminate gravity-wave measurements and distort measured flow spectra. In unfavorable mesospheric cases, those biases can reach several tens of

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meters per second. A systematic assessment of wind retrieval error is therefore essential for reliable dynamical inference.

The MF radar wind retrieval is therefore not only a geophysical measurement problem. It is also a signal processing problem built on a physical backscatter model. The signal model developed by Holdsworth provides a practical framework for representing scatterer distributions, anisotropy, and the resulting echo evolution in SA radar systems [19]. In the MF radar practice, wind retrieval is conventionally carried out with FCA [20], [21]. Existing MF radar studies have mainly examined the physical interpretation of the echoes, the implementation of FCA, the characterization of FCA biases, and the improvement of correlation-parameter extraction under noisy conditions [20], [21], [22], [23]. This literature is important because it directly informs practical MF radar wind retrieval. However, the conditional accuracy benchmark attainable from the full-MF radar observations under a specified echo model and scattering realization has not been sufficiently characterized. This has led the community to follow two practical routes. One is to characterize the error statistically. The other is to correct it empirically through calibration and intercomparison.

In the MF radar literature, the first route is statistical characterization of wind retrieval error. It has mainly taken two related forms. These are: 1) documenting systematic bias, especially the underestimation produced by conventional FCA and 2) quantifying retrieval error through variance, uncertainty, or mismatch statistics. For the first quantity, Reid's review of MF/HF radar techniques identified underestimation of wind magnitude as the major practical deficiency of FCA [21]. For small ideally configured MF/HF systems, the underestimation is typically on the order of 15 % to 40 %, while wind direction is usually much better preserved. Direct radar intercomparisons show that this bias is also height dependent. Reid et al. [24] found that MF partial reflection radar winds are underestimated relative to meteor radar winds. They further reported that the underestimation increases from 80 to 98 km, with Buckland Park MF underestimation of about 10%–20% near 86 km and much stronger degradation aloft. Under realistic horizontal wind gradients, Sürücü et al. [25] likewise showed that systematic errors arise naturally in SA FCA estimators under inhomogeneous flow conditions. For the second quantity, Gibbins and Kavanagh [17] showed that, for Antarctic MF radars using FCA, the width of the wind distribution and the occurrence of outliers are strongly linked to anisotropy and changing ionization conditions. Previous theoretical studies have already established important performance characteristics of spaced-antenna and correlation-based wind estimators. Hassenpflug et al. [26] showed that the variance of FCA wind estimates in SA systems depends strongly on anisotropic scattering and baseline orientation. Doviak et al. [27] compared several SA wind estimators theoretically and by simulation, and showed that estimator error depends on system configuration and lag spacing rather than on a single reduced correlation coefficient alone. In the space-physics literature, the first quantity is usually discussed as bias or underestimation. The second is usually discussed as retrieval error, variance, uncertainty, or mismatch statistics. In signal processing terms, the first corresponds to estimator bias. The second corresponds to variance and, more generally, to mse. For a potentially biased estimator, the mean-square error contains both variance and squared-bias contributions. The conventional CRB considered below bounds the covariance of an unbiased or locally unbiased estimator and therefore should be compared directly with estimator dispersion only when the bias contribution is negligible. We explicitly test this condition numerically in Section IV by reporting the Monte Carlo bias, standard deviation, and RMSE of both the conditional MLE and conventional FCA. This distinction allows the variance-based CRB comparison and the total retrieval error to be interpreted separately. Zhang et al. [28] further analyzed the performance of correlation estimators in the presence of receiver noise. The cited studies provide theoretical variance or precision expressions for specified spaced-antenna and correlation-based estimators under their respective statistical assumptions. Such expressions characterize the performance of a particular processing rule. The present work addresses a complementary quantity: the Fisher information and covariance lower bound implied by the adopted full-MF radar echo likelihood under a fixed and known scattering realization Ξ. Within this conditional model and the applicable unbiasedness conditions, the bound does not depend on selecting a particular FCA implementation. This distinction does not imply weaker assumptions or broader applicability: knowing the particular scattering realization is a stronger auxiliary-information condition than that available to conventional FCA. We therefore use the bound as an idealized conditional reference, not as a replacement for previous FCA error theories or as a direct measure of processing loss. Its additional role is to quantify how accurately a local full-signal correction must be initialized to approach the conditional information benchmark, as examined through the signal-to-noise ratio (SNR)–reference-mismatch results in Section IV.

Another practical route is empirical correction through cross-instrument validation, calibration, and data homogenization. In the space physics and remote sensing literature, this route remains dominant. Reid et al. [24] explicitly emphasized that intercomparisons are mainly used for validation and calibration. They also noted that such studies help clarify the relative strengths and weaknesses of different techniques. The same study pointed out that more recent MF partial reflection work often calibrates MF results against meteor radars. Wilhelm et al. [29] compared 11 years of MF radar and meteor radar winds in northern Norway. They used the meteor radar as the reference, estimated empirical correction factors in the overlap region, and then constructed a consistent and homogeneous 60–110-km wind data product. Similar cross-instrument practice also appears in lidar comparisons, where MF radar winds are assessed on matched time-height grids to diagnose systematic disagreement and improve measurement interpretation [12], [30]. These calibration- and intercomparison-based approaches are operationally valuable and often unavoidable for long-term geophysical data products. However, they rely on external reference instruments by construction. They quantify relative disagreement between instruments rather than the intrinsic optimum that conventional FCA processing in

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MF radar could achieve under a specified signal model. They also do not directly reveal whether the dominant performance loss originates from correlation-parameter extraction, from the nonlinear FCA mapping itself, or from a more fundamental gap between FCA and the model-based optimum estimator.

These calibration- and intercomparison-based approaches that rely on external reference instruments still leave a different question unanswered. That question arises from the standpoint of array radar signal processing. In that field, estimator error is commonly analyzed through signal models and information-theoretic tools rather than through cross-instrument validation alone. The issue is therefore posed in a different form. The question is not only how large the disagreement between instruments is. It is also what error level is implied by the assumed signal model and how that error depends on system design. Because both SA radar and multiple-input–multiple-output (MIMO) radar are multiantenna architectures, the MIMO literature provides a useful reference point even though the underlying targets and scattering assumptions are different. Fishler et al. [31] showed that spatial diversity can fundamentally improve radar performance relative to conventional coherent array processing. He et al. [32] analyzed noncoherent MIMO radar for joint location and velocity estimation. They derived closed-form CRBs and showed that increasing the number of channels can reduce the mean squared error (mse) and lower the threshold above which the estimator closely follows the CRB. Cao et al. [33] likewise derived maximum-likelihood-based velocity estimators and CRBs for distributed MIMO radar. They showed that velocity estimation performance improves with increasing antenna number and can approach the CRB under favorable conditions. In atmospheric applications, a related motivation is already visible in multistatic meteor radar work, where improved spatial coverage is accompanied by geometry-dependent velocity errors that must be quantified explicitly [34]. Hocking showed that some multistatic configurations can incur uncertainties large enough to make certain spatial regions practically unusable [34]. Taken together, these studies show that signal processing analysis can yield quantitative error expressions. They also provide practical guidance for system configuration and algorithm design.

Such results are possible because signal processing analysis starts from an explicit model and from the Fisher information implied by that model. This framework makes it possible to predict attainable accuracy, study scaling laws, and identify dominant error sources without first building a complete physical radar system. For MF radar, this capability is especially important. The practical objective of the present work is to characterize the sensing capability of MF radar by the mse. In many cases, however, one cannot build multiple candidate radar systems simply to determine that limit empirically. A model-based analysis using the Fisher information and the associated CRB is therefore one of the most effective ways to predict the achievable performance limit in advance.

Another modeling choice should be stated before the estimation questions are posed. The ultimate goal of this work is to characterize MF radar wind retrieval capability by the mse. However, the real MLT wind field is modulated by tides, gravity waves, turbulence, and other spatiotemporal inhomogeneities within the radar sampling volume. If all of this variability is retained from the outset, the resulting echo inversion becomes strongly nonlinear. It does not readily yield a transparent closed-form performance analysis. In contrast, conventional MLT wind products are usually formed as mean winds over finite retrieval windows and finite observation volumes. For example, monostatic meteor radar analyses commonly report mean horizontal winds over areas with an approximately 200–300-km radius. They also use 1–2-h temporal bins and 2–3-km vertical bins [35]. Recent instrument-comparison work likewise treats 1 h-cadence wind products as spatial averages over the observation region. It also notes explicitly that the uniform-wind approximation is only a local idealization [36]. Here, we make this local averaging assumption explicit and adopt a homogeneous horizontal wind field as an idealized short-window, limited-volume reference model. This choice does not imply that real MLT winds are exactly homogeneous. Instead, it provides a tractable closed-form benchmark from which the estimation capability of conventional FCA retrieval in MF radar can be quantified before more general inhomogeneous extensions are considered.

A second conditioning assumption is required for the statistical performance analysis. We use Ξ to denote the prescribed scattering realization and nonwind model/system inputs required to construct the adopted noiseless echo template. In the present formulation, the horizontal wind vector $\mathbf{v} = [v_x, v_y]^{\mathrm{T}}$ is the only unknown deterministic parameter of interest. The quantities contained in Ξ are conditioned upon and treated as fixed and known, whereas the receiver noise is random, and its covariance is specified by the adopted statistical model. No additional unknown nuisance parameter is estimated. The CRB developed below is therefore a conditional, model-specific covariance bound under this prescribed information condition rather than an unconditional bound obtained by marginalizing over unknown stochastic atmospheric scattering. Estimation of Ξ from measured echoes and analysis of uncertainty or mismatch in the prescribed scattering information are outside the scope of the present work.

Against this background, the present work focuses on three questions that remain insufficiently clarified within the adopted MF radar echo framework.

1) What conditional Fisher information about the homogeneous horizontal wind vector is contained in the complete MF radar echo observations under a fixed and known scattering realization Ξ?
2) Can a conditional full-signal numerical MLE approach the resulting CRB at high SNR, and how does conventional FCA compare with this benchmark under the prescribed simulation setting?
3) How does the reference-wind accuracy required for a locally linearized full-signal correction to remain near the conditional benchmark depend on SNR, and how should improvement relative to FCA be distinguished from actual attainment of that bound?

This article addresses these questions within a unified conditional full-signal framework for homogeneous horizontal wind retrieval. Specifically, it examines the Fisher-information

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benchmark available from the complete MF radar echo observations under a fixed scattering realization, evaluates how conventional FCA performs relative to this benchmark, and identifies processing mechanisms that contribute to the observed performance margin. The main contributions are summarized as follows.

1) *Conditional Full-Signal Benchmark:* We derive an analytical Fisher-information matrix (FIM) and CRB for horizontal wind estimation from the adopted MF radar echo likelihood under a fixed and known scattering realization Ξ. The bound applies under the corresponding regularity and unbiasedness conditions and complements existing error analyses of specified FCA/spaced-antenna estimators.
2) *Numerical Tightness and Comparison:* We use a conditional MLE with the same known Ξ to examine the high-SNR tightness of the bound and evaluate conventional FCA relative to this conditional benchmark. The comparison does not isolate processing loss because the benchmark assumes additional scattering-realization knowledge.
3) *Reference Accuracy for Local Correction:* We quantify how the reference-wind accuracy required for a locally linearized full-signal correction to remain near the conditional CRB depends on SNR. The results distinguish a reduction in FCA error from attainment of the bound; the correction continues to require the known scattering template.

II. FCA Wind Estimation

This section summarizes the conventional FCA processing chain used as the comparison estimator in the subsequent full-signal performance analysis. The purpose is to establish the FCA quantities required for the later comparison, rather than to present the FCA estimator itself as a new contribution. According to the principles of FCA, the horizontal atmospheric wind field components are expressed as follows:

$$v_x = \frac{hg - bf}{ab - h^2} \tag{1}$$

$$v_y = \frac{hf - ag}{ab - h^2} \tag{2}$$

where a, b, f, g, and h are the characteristic ellipsoid parameters. The linear coefficients f and g satisfy

$$\tau'_{nn'} = -f\xi_{nn'} - g\eta_{nn'}. \tag{3}$$

In (3), $\tau'_{nn'}$ represents the time lag corresponding to the peak of the cross-correlation function between the time series of the nth and n'th receiver channels, which is obtained by

$$\frac{\partial \rho_{nn'}(\tau)}{\partial \tau} = 0 \tag{4}$$

$\xi_{nn'}$ and $\eta_{nn'}$ represent the spatial baseline separations between the nth and n'th receiving antennas along the x-axis and y-axis, respectively, i.e.,

$$\xi_{nn'} = x^{\mathrm{r}}_n - x^{\mathrm{r}}_{n'} \tag{5}$$

$$\eta_{nn'} = y^{\mathrm{r}}_n - y^{\mathrm{r}}_{n'}. \tag{6}$$

With N receiving antennas, $N_s = N(N-1)/2$ pairwise lag equations can be formed. However, the number of antenna pairs alone does not guarantee identifiability of f and g. Define the baseline design matrix $\mathbf{B} = [-\boldsymbol{\xi}, -\boldsymbol{\eta}] \in \mathrm{R}^{N_s \times 2}$. The two coefficients f and g are uniquely identifiable in the least-squares sense only if $\mathrm{rank}(\mathbf{B}) = 2$. Therefore, two receiving antennas are insufficient because they provide only one baseline equation. At least three receivers with a noncollinear planar geometry are required to provide two independent baseline directions. Define

$$\boldsymbol{\xi} = \left[\xi_{12}, \ldots, \xi_{nn'}, \ldots, \xi_{N1}\right]^{\dagger} \tag{7}$$

$$\boldsymbol{\eta} = \left[\eta_{12}, \ldots, \eta_{nn'}, \ldots, \eta_{N1}\right]^{\dagger} \tag{8}$$

$$\boldsymbol{\tau}' = \left[\tau'_{12}, \ldots, \tau'_{nn'}, \ldots, \tau'_{N1}\right]^{\dagger}. \tag{9}$$

Stacking all pairwise equations in (3) yields

$$\boldsymbol{\tau}' = \left[-\boldsymbol{\xi}, -\boldsymbol{\eta}\right] \begin{bmatrix} f \\ g \end{bmatrix}. \tag{10}$$

Hence, the least-squares estimate of $[f, g]^{\dagger}$ is

$$\begin{bmatrix} f \\ g \end{bmatrix} = \left[-\boldsymbol{\xi}, -\boldsymbol{\eta}\right]^{+} \boldsymbol{\tau}' \tag{11}$$

where $(\cdot)^{+}$ denotes the pseudoinverse. When $\mathrm{rank}(\mathbf{B}) = 2$, the pseudoinverse solution in (11) is the unique least-squares estimate of f and g. If $\mathrm{rank}(\mathbf{B}) < 2$, the individual coefficients are not identifiable; although the Moore–Penrose pseudoinverse still returns a minimum-norm numerical solution, different parameter vectors can produce the same observable baseline-lag combination. The remaining FCA shape parameters are obtained from the decorrelation-time relation

$$\left(\tau''_{nn'}\right)^2 = a\xi^2_{nn'} + b\eta^2_{nn'} + h\xi_{nn'}\eta_{nn'}. \tag{12}$$

Similarly, unique identification of the three shape parameters a, b, and h requires the corresponding quadratic baseline design matrix to have full column rank. Denoting this matrix by $\mathbf{D}$, the identifiability condition is $\mathrm{rank}(\mathbf{D}) = 3$. This requires at least three suitably oriented baselines, but the number of baselines alone is not sufficient. The total correlation width is related to the peak lag and the decorrelation time through

$$\tau^2_{nn'} = \left(\tau'_{nn'}\right)^2 + \left(\tau''_{nn'}\right)^2 \tag{13}$$

where $\tau''_{nn'}$ denotes the characteristic decorrelation time associated with the (n, n') baseline. Fig. 1 illustrates the two correlation parameters used by FCA. Define

$$\left(\boldsymbol{\tau}''\right)^2 = \left[\left(\tau''_{12}\right)^2, \ldots, \left(\tau''_{nn'}\right)^2, \ldots, \left(\tau''_{N1}\right)^2\right]^{\dagger} \tag{14}$$

$$\boldsymbol{\xi}^2 = \left[\xi^2_{12}, \ldots, \xi^2_{nn'}, \ldots, \xi^2_{N1}\right]^{\dagger} \tag{15}$$

$$\boldsymbol{\eta}^2 = \left[\eta^2_{12}, \ldots, \eta^2_{nn'}, \ldots, \eta^2_{N1}\right]^{\dagger} \tag{16}$$

$$\boldsymbol{\xi\eta} = \left[\xi_{12}\eta_{12}, \ldots, \xi_{nn'}\eta_{nn'}, \ldots, \xi_{N1}\eta_{N1}\right]^{\dagger}. \tag{17}$$

Stacking (12) over all antenna pairs yields

$$\left(\boldsymbol{\tau}''\right)^2 = \left[\boldsymbol{\xi}^2, \boldsymbol{\eta}^2, \boldsymbol{\xi\eta}\right] \begin{bmatrix} a \\ b \\ h \end{bmatrix}. \tag{18}$$

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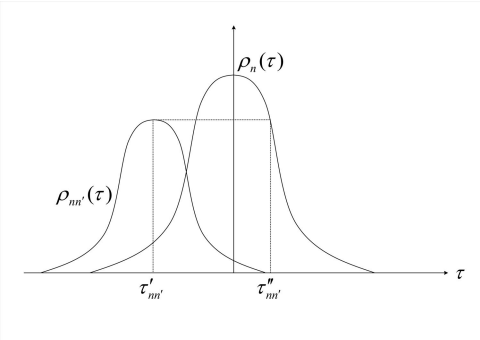


Fig. 1. Correlation parameters $\tau'_{nn'}$ and $\tau''_{nn'}$.

Therefore, the least-squares estimate of $[a, b, h]^\dagger$ is

$$\begin{bmatrix} a \\ b \\ h \end{bmatrix} = \left[\boldsymbol{\xi}^2, \boldsymbol{\eta}^2, \boldsymbol{\xi}\boldsymbol{\eta}\right]^+ \left(\boldsymbol{\tau}''\right)^2. \tag{19}$$

First, $\tau'_{nn'}$ is extracted from the peak of the sampled cross correlation. Let $f_n[k]$ be the received sample of the nth channel. The normalized cross correlation between channels n and n' is defined as follows:

$$\rho_{nn'}(\tau_l) = \frac{\sum_{k=1}^{K-|l|} f_n[k]\, f_{n'}{}^*(k+l)}{(K-|l|)\left(\sum_{k=1}^{K}|f_n[k]|^2 \sum_{k=1}^{K}|f_{n'}[k]|^2\right)^{1/2}} \tag{20}$$

where $\tau_l = lT_s$, $l \in \mathcal{I}_l = \{-(K-1), \ldots, K-1\}$, $(\cdot)^*$ denotes the complex conjugation, and K is the number of snapshots. For the analytical development below, we introduce a local second-order representation of the sampled correlation peak. Specifically, the logarithm of the peak magnitude is approximated locally by a quadratic model over a symmetric neighborhood. This representation provides an analytically tractable description of perturbations in the extracted FCA correlation parameters and is used in the subsequent local error analysis. It should be distinguished from the numerical correlation-fitting procedure used in Section IV, where the actual FCA implementation employs an eighth-order polynomial fit

$$\hat{l}_{nn'} = \arg\max_{l\in\mathcal{I}_l} |\rho_{nn'}(\tau_l)|. \tag{21}$$

The quadratic representation above is introduced for local analytical tractability and should not be interpreted as prescribing the polynomial order used in the numerical FCA processor. In the simulations, the intermediate correlation quantities are obtained using the fixed eighth-order fitting procedure described in Section IV, and the resulting quantities are subsequently mapped to the FCA wind estimate

$$\ln\left|\rho_{nn'}\left(\tau_{\hat{l}_{nn'}+m}\right)\right| \approx c_{0,nn'} + c_{1,nn'} mT_s + c_{2,nn'}(mT_s)^2. \tag{22}$$

For the local analytical error formulation, the sampled correlation peak is represented by a second-order approximation to its logarithmic magnitude. The resulting peak-lag and decorrelation-time estimates are denoted by $\hat{\tau}'$ and $(\hat{\tau}'')^2$, respectively. The detailed quadratic fitting and least-squares expressions are provided in Appendix A. This analytical surrogate should be distinguished from the fixed eighth-order polynomial fitting procedure used in the numerical FCA implementation of Section IV.

Finally, the FCA wind estimates are

$$\hat{v}_{x,\mathrm{FCA}} = \frac{\hat{h}\hat{g} - \hat{b}\hat{f}}{\hat{a}\hat{b} - \hat{h}^2} \tag{23}$$

$$\hat{v}_{y,\mathrm{FCA}} = \frac{\hat{h}\hat{f} - \hat{a}\hat{g}}{\hat{a}\hat{b} - \hat{h}^2}. \tag{24}$$

Equations (23) and (24) reveal the main limitation of the traditional FCA estimator. Both components are nonlinear rational functions of the extracted correlation parameters. Consequently, peak-location errors, local-fitting bias, and array-geometry ill-conditioning are all amplified through the denominator, which explains why FCA is generally suboptimal with respect to the CRB.

To analyze the FCA error, it is useful to make the propagation mechanism explicit. Define the intermediate parameter vector $\boldsymbol{\theta}_{\mathrm{FCA}} = [a, b, h, f, g]^\dagger$ and its estimate $\widehat{\boldsymbol{\theta}}_{\mathrm{FCA}} = [\hat{a}, \hat{b}, \hat{h}, \hat{f}, \hat{g}]^\dagger$. Let $\boldsymbol{v}_{\mathrm{FCA}}(\boldsymbol{\theta}_{\mathrm{FCA}}) = [v_{x,\mathrm{FCA}}, v_{y,\mathrm{FCA}}]^\dagger$ denote the nonlinear mapping induced by (1) and (2). A first-order perturbation expansion around the true intermediate parameters gives

$$\widehat{\boldsymbol{v}}_{\mathrm{FCA}} - \boldsymbol{v} \approx \boldsymbol{G}_{\mathrm{FCA}}\left(\widehat{\boldsymbol{\theta}}_{\mathrm{FCA}} - \boldsymbol{\theta}_{\mathrm{FCA}}\right) \tag{25}$$

where $\boldsymbol{G}_{\mathrm{FCA}} = \partial \boldsymbol{v}_{\mathrm{FCA}}(\boldsymbol{\theta})/\partial \boldsymbol{\theta}^\dagger\big|_{\boldsymbol{\theta}=\boldsymbol{\theta}_{\mathrm{FCA}}}$ is the Jacobian of the FCA nonlinear mapping. Let $b_{\mathrm{FCA}} = E[\hat{v}_{\mathrm{FCA}}] - v$ be the FCA bias vector. Under a local linearization and when the bias is negligible, the covariance associated with FCA fluctuations can be approximated by

$$\boldsymbol{C}_{\mathrm{FCA}} = \mathrm{E}\left\{\left(\widehat{\boldsymbol{v}}_{\mathrm{FCA}} - \boldsymbol{v}\right)\left(\widehat{\boldsymbol{v}}_{\mathrm{FCA}} - \boldsymbol{v}\right)^\dagger\right\} \approx \boldsymbol{G}_{\mathrm{FCA}}\boldsymbol{C}_\theta\boldsymbol{G}_{\mathrm{FCA}}^\dagger \tag{26}$$

where $\boldsymbol{C}_\theta = \mathrm{E}\{(\widehat{\boldsymbol{\theta}}_{\mathrm{FCA}} - \boldsymbol{\theta}_{\mathrm{FCA}})(\widehat{\boldsymbol{\theta}}_{\mathrm{FCA}} - \boldsymbol{\theta}_{\mathrm{FCA}})^\dagger\}$ is determined by the errors in the extracted correlation parameters. For a biased FCA estimate, however, the total error second moment additionally contains the bias contribution $b_{\mathrm{FCA}}b_{\mathrm{FCA}}^\dagger$. Therefore, (26) should be interpreted as a local covariance-propagation model rather than as a complete expression for the FCA mse when the bias is nonnegligible. Equation (26) shows that, within the local approximation, the FCA covariance is governed jointly by the uncertainty of the extracted correlation parameters and by the conditioning of the nonlinear mapping from (a, b, h, f, g) to (v_x, v_y). These terms characterize estimator-level error propagation within FCA. However, the conditional full-signal benchmark additionally assumes knowledge of the prescribed scattering realization Ξ, which is not supplied to FCA. Therefore, (26) cannot determine how much of the complete FCA–CRB separation is attributable to FCA processing itself, and it should not be interpreted as an exact or exhaustive decomposition of that separation. The additional Monte Carlo analysis in Section IV shows that conventional FCA may also contain a significant bias component, which must be included when total mse is considered.

III. Full-Signal Conditional Performance Analysis

A. Echo Signal Model

Consider a 3-D Cartesian coordinate system (x, y, z) with the origin as the reference point. Here, the x-, y-, and

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z-axes denote the zonal, meridional, and vertical directions, respectively. Assume an MF radar system employing the SA configuration, where the $n(n = 1, \ldots, N)$ receiving antenna is located at $\boldsymbol{d}_n = [x_n^{\mathrm{r}}, y_n^{\mathrm{r}}, 0]^{\dagger}$. Based on the atmospheric backscatter model proposed by Holdsworth, N_{sca} scatterers are assumed uniformly distributed within the radar detection volume, whose zonal, meridional, and vertical supports are denoted by $r_{i,x} \in [x_{\min}, x_{\max}]$, $r_{i,y} \in [y_{\min}, y_{\max}]$, and $r_{i,z} \in [z_{\min}, z_{\max}]$, respectively. At the $k(k = 1, \ldots, K)$ sampling instant, the position of the $i(i = 1, \ldots, N_{\mathrm{sca}})$ scatterer is given by

$$\boldsymbol{r}_i[k] = \boldsymbol{r}_{i,0} + \left[v_x k T_{\mathrm{s}}, v_y k T_{\mathrm{s}}, 0\right]^{\dagger} \tag{27}$$

where $\boldsymbol{r}_{i,0} = [r_{i,x}, r_{i,y}, r_{i,z}]^{\dagger}$ denotes the initial position of the scatterer. For the conditional analysis, the horizontal wind vector $\mathbf{v} = [v_x, v_y]^{\mathrm{T}}$ is the only unknown deterministic parameter of interest. We use Ξ to collect the prescribed quantities required to construct the adopted noiseless echo template apart from $\mathbf{v}$. These include the number and initial positions of the scatterers, the antenna geometry, the wavelength and sampling configuration, and the nonwind parameters retained in the amplitude and propagation-phase expressions. In the adopted simulation model, all scatterers use the same prescribed scattering coefficient; this common coefficient is absorbed into the overall amplitude normalization. These quantities are treated as fixed and known when evaluating the conditional likelihood, MLE, and CRB; they are not estimated from the observations in the present analysis. This conditioning does not imply that the echo amplitude or propagation phase is constant. The scatterer positions evolve according to (27) and the corresponding amplitude and phase terms are evaluated at these moving positions. Thus, their values generally depend on the candidate wind vector even though the prescribed nonwind template inputs remain fixed. The receiver noise introduced below is the random component of the repeated-noise experiment, and its covariance is specified by the adopted statistical model. No additional unknown nuisance parameter is included in the present conditional formulation. The statistical roles of the quantities involved in the conditional analysis are summarized in Table I. The fixed-and-known Ξ condition therefore defines an idealized information benchmark; it does not assert that the microscopic scattering realization is available in operational MF radar measurements. Estimation of Ξ, marginalization over random atmospheric scattering, and uncertainty or mismatch in the prescribed scattering information are outside the present formulation.

The support limits $x_{\min}$, $x_{\max}$, $y_{\min}$, $y_{\max}$, $z_{\min}$, and $z_{\max}$ are kept symbolic in the model derivation and will be assigned numerical values only in the simulation section. Here, the homogeneous horizontal wind field is interpreted as a local zero-order approximation to the mean flow within one retrieval window and over the effective radar sampling volume. This idealization is introduced so that the performance of conventional FCA retrieval in MF radar can be analyzed in closed form, not because the real MLT wind field is assumed to be free of tides, gravity waves, or turbulence [35], [36]. Accordingly, the horizontal wind vector is written as $[v_x, v_y, 0]^{\dagger}$, where v_x and v_y are treated as constants within the adopted local model. Accordingly, this assumption defines a local idealized retrieval scenario and should not be interpreted as requiring a globally stationary or physically quiescent atmosphere. In addition, T_{s} represents the radar sampling interval satisfying the Nyquist sampling rate. The echo signal received by the nth antenna from the $i(i = 1, \ldots, N_{\mathrm{sca}})$ scatterer at the kth sampling instant is expressed as follows:

$$y_{n,i}[k] = \alpha_n(\boldsymbol{r}_i[k]) \exp\{-j\phi_n(\boldsymbol{r}_i[k])\} \tag{28}$$

where $j = \sqrt{-1}$ is the imaginary unit, and $\alpha_n(\boldsymbol{r}_i[k])$ and $\phi_n(\boldsymbol{r}_i[k])$ denote the amplitude and phase of the received signal, respectively, both of which are functions of the scatterer's position.

The detailed transmit-pattern, aspect-sensitivity, range-gate, amplitude, and propagation-phase expressions leading to the adopted echo model are provided in Appendix B.

The noiseless echo received by the nth radar antenna at the kth sampling instant is the superposition of contributions from all scatterers

$$y_n[k] = \sum_{i=1}^{N_{\mathrm{sca}}} \alpha_n(\boldsymbol{r}_i[k]) \exp\{-j\phi_n(\boldsymbol{r}_i[k])\}. \tag{29}$$

Substituting the simplified amplitude and phase expressions into (29) yields

$$\begin{aligned} y_n[k] &= \sum_{i=1}^{N_{\mathrm{sca}}} \alpha_n(\boldsymbol{r}_i(kT_{\mathrm{s}})) \exp\{-j\phi_n(\boldsymbol{r}_i(kT_{\mathrm{s}}))\} \\ &= \sum_{i=1}^{N_{\mathrm{sca}}} \left\{\exp\left\{-|\boldsymbol{r}_i(kT_{\mathrm{s}})|^2\right\}\right\}^{1/2} \\ &\quad \times \exp\left\{-j\left(|\boldsymbol{r}_i(kT_{\mathrm{s}})|^2 + |\boldsymbol{r}_i(kT_{\mathrm{s}}) - \boldsymbol{d}_n|^2\right)\right\}. \end{aligned} \tag{30}$$

Given that only the horizontal wind field is estimated, the z-direction does not contain unknown velocity components.

TABLE I
STATISTICAL ROLES OF THE QUANTITIES IN THE CONDITIONAL ANALYSIS

Quantity	Statistical role in the conditional analysis
Horizontal wind $\mathbf{v} = [v_x, v_y]^{\mathrm{T}}$	Unknown deterministic parameter of interest; the only parameter estimated in the present formulation.
Scatterer number and initial positions $N_{\mathrm{sca}}, \{r_{i,0}\}_{i=1}^{N_{\mathrm{sca}}}$	Prescribed scattering realization; fixed and known in conditional template construction. One generated realization is retained across repeated-noise trials at a given setting.
Common scattering coefficient p_0 and overall amplitude normalization	Prescribed and identical for all scatterers in the adopted simulation model; the common coefficient is absorbed into the overall amplitude normalization.
Antenna geometry, wavelength, sampling interval, and observation indices	Known system and sampling specifications.
Beam-pattern, aspect-sensitivity, range-gate, and propagation-phase model inputs	Prescribed non-wind model inputs used in constructing the conditional template.
Receiver noise $\mathbf{w}$	Random, zero-mean circular complex Gaussian noise; independent realizations are regenerated across Monte Carlo trials.
Noise covariance $\mathbf{Q} = \sigma_w^2\mathbf{I}$	Specified in the conditional statistical model and independent of the candidate wind vector.
Unknown nuisance parameters	None are included in the present conditional formulation; unknown scattering, calibration, or noise parameters would require an extended model.

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Define the parameter vector $\boldsymbol{v} = (v_x, v_y)^\dagger$. Then, (30) can be rewritten as follows:

$$\begin{aligned} y_n[k] &= y_n(kT_s; \mathbf{v}, \Xi) \\ &= \sum_{i=1}^{N_{sca}} \exp\left\{-\frac{A(i,k,\boldsymbol{v})}{2}\right\} \\ &\quad \times \exp\{-j(A(\boldsymbol{v},i,k) + B(\boldsymbol{v},i,k,n))\} \\ &= \sum_{i=1}^{N_{sca}} \exp\left\{-\left(\frac{1}{2}+j\right)A(\boldsymbol{v},i,k) - jB(\boldsymbol{v},i,k,n)\right\} \end{aligned} \quad (31)$$

where

$$A(\boldsymbol{v},i,k) = \left(r_{i,x} + v_x kT_s\right)^2 + \left(r_{i,y} + v_y kT_s\right)^2 \quad (32)$$

$$B(\boldsymbol{v},i,k,n) = \left(r_{i,x} + v_x kT_s - x_n^r\right)^2 + \left(r_{i,y} + v_y kT_s - y_n^r\right)^2. \quad (33)$$

After K snapshots, the time series recorded by the nth receiver channel is

$$\boldsymbol{f}_n = \left[f_n[1], \ldots, f_n[K]\right]^\dagger = \boldsymbol{y}_n + \boldsymbol{w}_n \quad (34)$$

where

$$f_n[k] = y_n[k] + w_n[k] \quad (35)$$

$$\boldsymbol{y}_n = \left[y_n[1], \ldots, y_n[K]\right]^\dagger \quad (36)$$

where $w_n[k]$ denotes additive complex white Gaussian noise (AWGN) at the receiver. Define

$$\boldsymbol{w}_n = [w_n[1], \ldots, w_n[K]]^\dagger \quad (37)$$

and assume $\boldsymbol{w}_n \sim \mathcal{CN}(\mathbf{0}, \sigma_w^2 \boldsymbol{I}_K)$. Stacking the observations from all receiver channels into a single long vector yields

$$\boldsymbol{f} = \left[\boldsymbol{f}_1^\dagger, \ldots, \boldsymbol{f}_N^\dagger\right]^\dagger = \boldsymbol{y}(\mathbf{v}; \Xi) + \mathbf{w} \quad (38)$$

for notational simplicity, the dependence on Ξ is suppressed in intermediate expressions when no ambiguity arises. Where

$$\boldsymbol{y} = \left[\boldsymbol{y}_1^\dagger, \ldots, \boldsymbol{y}_N^\dagger\right]^\dagger \quad (39)$$

$$\boldsymbol{w} = \left[\boldsymbol{w}_1^\dagger, \ldots, \boldsymbol{w}_N^\dagger\right]^\dagger. \quad (40)$$

B. Conditional CRB for Wind Field Parameter Estimation

The FCA method retrieves the wind velocity vector $\boldsymbol{v}$ by exploiting the spatiotemporal correlations among different receiver channels to estimate time lags. This section derives the CRB for the covariance matrix of any unbiased estimator of $\boldsymbol{v}$ under the signal model in (38).

Based on (38), the observation vector $\boldsymbol{f}$ follows a complex normal distribution $\mathcal{CN}(\boldsymbol{y}(\boldsymbol{v}; \Xi), \boldsymbol{Q})$, where

$$\boldsymbol{Q} = \sigma_w^2 \boldsymbol{I}_{KN}. \quad (41)$$

The covariance $\mathbf{Q}$ is treated as specified and independent of the candidate wind vector in the conditional likelihood and Fisher-information calculation. In the simulations, σ_w^2 is set according to the stated SNR for the prescribed signal realization. The corresponding probability density function is

$$p(\boldsymbol{f}; \boldsymbol{v}|\Xi) = \frac{1}{\pi^{KN}\det(\boldsymbol{Q})} \exp\left\{-(\boldsymbol{f}-\boldsymbol{y})^\dagger \boldsymbol{Q}^{-1}(\boldsymbol{f}-\boldsymbol{y})\right\}. \quad (42)$$

Taking the natural logarithm of (42) yields the log-likelihood function

$$\ln p(\boldsymbol{f}; \boldsymbol{v}|\Xi) = -\ln\left(\pi^{KN}\det(\boldsymbol{Q})\right) - (\boldsymbol{f}-\boldsymbol{y})^\dagger \boldsymbol{Q}^{-1}(\boldsymbol{f}-\boldsymbol{y}). \quad (43)$$

Since $\boldsymbol{Q}$ is independent of $\boldsymbol{v}$, the constant term does not affect the Fisher-information. For convenience, define the log-likelihood as follows:

$$\begin{aligned} \mathcal{L}(\boldsymbol{f}; \boldsymbol{v}|\Xi) &= \ln p(\boldsymbol{f}; \boldsymbol{v}) \\ &= -(\boldsymbol{f}-\boldsymbol{y})^\dagger \boldsymbol{Q}^{-1}(\boldsymbol{f}-\boldsymbol{y}) - \ln\left(\pi^{KN}\det(\boldsymbol{Q})\right). \end{aligned} \quad (44)$$

With $\boldsymbol{Q}^{-1} = \sigma_w^{-2}\boldsymbol{I}_{KN}$, the quadratic term in (44) is the squared Euclidean mismatch normalized by the noise power.

The corresponding MLE is therefore defined by

$$\begin{aligned} \widehat{\boldsymbol{v}}_{ML} &= \arg\max_{\boldsymbol{v}} \mathcal{L}(\boldsymbol{f}; \boldsymbol{v}|\Xi) \\ &= \arg\min_{\boldsymbol{v}} (\boldsymbol{f} - \boldsymbol{y}(\boldsymbol{v}))^\dagger \boldsymbol{Q}^{-1}(\boldsymbol{f} - \boldsymbol{y}(\boldsymbol{v})). \end{aligned} \quad (45)$$

The MLE defined here is therefore a conditional full-signal MLE in which the fixed scattering realization is used in constructing the noiseless signal template.

Because $\boldsymbol{y}(\boldsymbol{v})$ is nonlinear in $\boldsymbol{v}$, (45) does not admit a simple closed-form solution in general and must be solved numerically. Define bias as $b_{ML} = E[\hat{v}_{ML}] - v$, and real covariance is $C_{ML} = E[(\hat{v}_{ML} - E[\hat{v}_{ML}])(\hat{v}_{ML} - E[\hat{v}_{ML}])^\dagger]$. The corresponding error second-moment matrix satisfies

$$M_{ML} = C_{ML} + b_{ML} b_{ML}^\dagger. \quad (46)$$

Under the standard regularity conditions, the conditional MLE becomes asymptotically unbiased and efficient; hence, its bias approaches 0, and its covariance approaches the conditional CRB

$$b_{ML} \to 0, C_{ML} \to C_{CRB}. \quad (47)$$

The convergence behavior described by Equation (47) is examined numerically in Section IV.

For the complex Gaussian model, the FIM is computed using the standard Slepian–Bangs formula

$$[\boldsymbol{J}(\boldsymbol{v}|\Xi)]_{ij} = 2\Re\left[\left(\frac{\partial \boldsymbol{y}}{\partial v_i}\right)^\dagger \boldsymbol{Q}^{-1}\left(\frac{\partial \boldsymbol{y}}{\partial v_j}\right)\right]. \quad (48)$$

Defining the parameter vector as $\boldsymbol{v} = (v_x, v_y)^\dagger$, the FIM with respect to $\boldsymbol{v}$ has the following structure:

$$\begin{aligned} \boldsymbol{J}(\boldsymbol{v}) &= 2\begin{bmatrix} \mathbf{J}_{xx} & \mathbf{J}_{xy} \\ \mathbf{J}_{yx} & \mathbf{J}_{yy} \end{bmatrix} \\ \mathbf{J}_{ij} &= \Re\left[\left(\frac{\partial \boldsymbol{y}}{\partial v_i}\right)^\dagger \boldsymbol{Q}^{-1}\left(\frac{\partial \boldsymbol{y}}{\partial v_j}\right)\right], \quad v_i, v_j \in \{v_x, v_y\}. \end{aligned} \quad (49)$$

The analytical signal derivatives with respect to v_x and v_y, together with their explicit substitution into the conditional FIM, are given in Appendix B.

Therefore, the complete FIM is

$$\boldsymbol{J}(\boldsymbol{v}|\Xi) = \begin{bmatrix} [\boldsymbol{J}(\boldsymbol{v})]_{11} & [\boldsymbol{J}(\boldsymbol{v})]_{12} \\ [\boldsymbol{J}(\boldsymbol{v})]_{21} & [\boldsymbol{J}(\boldsymbol{v})]_{22} \end{bmatrix}. \quad (50)$$

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Under the standard regularity and unbiasedness conditions, and at parameter values for which the conditional FIM is nonsingular, the conditional covariance of an unbiased estimator satisfies

$$\mathrm{Cov}_{\mathbf{w}}\left(\hat{\mathbf{v}}|\Xi\right) \geq \boldsymbol{C}_{\mathrm{CRB}}\left(\mathbf{v}|\Xi\right) \tag{51}$$

where

$$\boldsymbol{C}_{\mathrm{CRB}}\left(\mathbf{v}|\boldsymbol{\Xi}\right) = \boldsymbol{J}^{-1}\left(\mathbf{v}|\Xi\right). \tag{52}$$

For an unbiased estimator under the same conditional model, taking the trace gives the corresponding conditional vector-mse lower bound

$$\boldsymbol{E}_{\mathrm{w}}\left[\|\hat{\mathbf{v}}-\mathbf{v}\|_2^2|\Xi\right] \geq \mathrm{tr}\left[\boldsymbol{C}_{\mathrm{CRB}}\left(\mathbf{v}|\Xi\right)\right]. \tag{53}$$

This is a conditional bound under the specified model and information condition. It should not be interpreted as a universal mse lower bound for arbitrary biased estimators. The diagonal entries of $\boldsymbol{C}_{\mathrm{CRB}}(\mathbf{v}|\Xi)$ provide the corresponding componentwise variance lower bounds

$$\mathrm{CRB}_{v_x} = \left[\boldsymbol{C}_{\mathrm{CRB}}\left(\mathbf{v}|\Xi\right)\right]_{11} \tag{54}$$

$$\mathrm{CRB}_{v_y} = \left[\boldsymbol{C}_{\mathrm{CRB}}\left(\mathbf{v}|\Xi\right)\right]_{22}. \tag{55}$$

The matrix $\boldsymbol{C}_{\mathrm{CRB}}(\mathbf{v}|\boldsymbol{\Xi})$ is therefore the conditional covariance lower bound for unbiased estimation of $\mathbf{v}$ from the full-observation vector under the adopted echo model and prescribed information condition. It should not be interpreted as an unconditional or universal MF radar wind retrieval limit in which the atmospheric scattering realization is unknown and marginalized. Rather, it provides a model-specific full-signal benchmark against which estimators evaluated under the same fixed realization can be compared.

The conditional full-signal benchmark assumes knowledge of the prescribed scattering realization Ξ. Conventional FCA uses the known array geometry, the correlation model, and correlation quantities estimated from the observations, but it is not supplied with the complete microscopic scattering realization used to construct the full-signal template. Consequently, generating the observations under the same Ξ does not imply that the compared methods operate with the same auxiliary information. The separation between FCA and the conditional CRB may therefore reflect both FCA processing effects and the additional scattering-realization knowledge assumed by the conditional benchmark. The present experiment does not isolate or quantify these two contributions. Accordingly, the complete FCA–CRB separation should neither be interpreted as a pure measure of information loss caused by correlation-based processing nor as an accuracy gain that could necessarily be realized by modifying FCA alone. The local FCA covariance-propagation relation in (26) characterizes how uncertainty in the extracted correlation parameters propagates through the nonlinear FCA mapping. It therefore identifies estimator-level mechanisms contributing to FCA error, but it does not constitute an exact or exhaustive decomposition of the complete FCA–CRB separation.

C. Local Linearization-Based Best Linear Unbiased Estimator (BLUE)

To establish the performance bound of linear estimators for the MF radar system, this section derives a locally linearized BLUE based on the Gauss–Markov theorem [37]. The goal is not to replace the nonlinear signal model, but to construct a locally optimal linear approximation around a reference wind vector.

Starting from the conditional full-signal model $\boldsymbol{f} = \mathbf{y}(\boldsymbol{v};\boldsymbol{\Xi}) + \mathbf{w}$, let $\boldsymbol{v}_{\mathrm{ref}}$ be a reference wind vector obtained from prior information or an initial FCA estimate. A first-order Taylor expansion of the conditional signal template around $\boldsymbol{v}_{\mathrm{ref}}$ gives

$$\mathbf{y}\left(\boldsymbol{v};\boldsymbol{\Xi}\right) \approx \mathbf{y}\left(\boldsymbol{v}_{\mathrm{ref}};\boldsymbol{\Xi}\right) + \mathbf{H}\left(\boldsymbol{v}_{\mathrm{ref}};\boldsymbol{\Xi}\right)\left(\boldsymbol{v}-\boldsymbol{v}_{\mathrm{ref}}\right) \tag{56}$$

where

$$\mathbf{H}\left(\mathbf{v}_{\mathrm{ref}};\boldsymbol{\Xi}\right) = \left[\frac{\partial \boldsymbol{y}}{\partial \mathbf{v}_{\mathbf{x}}}, \frac{\partial \boldsymbol{y}}{\partial \mathbf{v}_{\mathbf{y}}}\right]\Bigg|_{\boldsymbol{v}=\boldsymbol{v}_{\mathrm{ref}}}. \tag{57}$$

The dependence on the prescribed Ξ is suppressed below only for notational brevity. Define the wind perturbation and observation residual as $\Delta\boldsymbol{v} = \boldsymbol{v}-\boldsymbol{v}_{\mathrm{ref}}$, $\Delta\boldsymbol{f} = \boldsymbol{f}-\mathbf{y}(\mathbf{v}_{\mathrm{ref}};\boldsymbol{\Xi})$, respectively,

$$\Delta\boldsymbol{f} = \boldsymbol{H}\Delta\boldsymbol{v} + \boldsymbol{w} + \mathcal{O}\left(\|\Delta\boldsymbol{v}\|_2^2\right). \tag{58}$$

Neglecting the higher order truncation term reduces the problem to estimating the real-valued increment $\Delta\boldsymbol{v}$ from complex observations.

Since $\Delta\boldsymbol{f}$ and $\boldsymbol{H}$ are complex-valued, whereas $\Delta\boldsymbol{v}$ is real-valued, it is convenient to rewrite the model in the real domain. This gives

$$\begin{bmatrix}\Re\{\Delta\boldsymbol{f}\}\\ \Im\{\Delta\boldsymbol{f}\}\end{bmatrix} = \begin{bmatrix}\Re\{\boldsymbol{H}\}\\ \Im\{\boldsymbol{H}\}\end{bmatrix}\Delta\boldsymbol{v} + \begin{bmatrix}\Re\{\boldsymbol{w}\}\\ \Im\{\boldsymbol{w}\}\end{bmatrix}. \tag{59}$$

Let $\widetilde{\Delta\boldsymbol{f}} = [\Re\{\Delta\boldsymbol{f}\}^\dagger, \Im\{\Delta\boldsymbol{f}\}^\dagger]^\dagger$ and $\widetilde{\boldsymbol{H}} = [\Re\{\boldsymbol{H}\}^\dagger, \Im\{\boldsymbol{H}\}^\dagger]^\dagger$. Because $\boldsymbol{w} \sim \mathcal{CN}(\mathbf{0}, \sigma_w^2\boldsymbol{I}_{KN})$, the real-valued noise vector has covariance $\boldsymbol{C}_{\widetilde{w}} = \sigma_w^2\boldsymbol{I}_{2KN}/2$. The Gauss–Markov theorem then gives the BLUE in the real domain as follows:

$$\Delta\widehat{\boldsymbol{v}}_{\mathrm{BLUE}} = \left(\widetilde{\boldsymbol{H}}^\dagger\widetilde{\boldsymbol{H}}\right)^{-1}\widetilde{\boldsymbol{H}}^\dagger\widetilde{\Delta\boldsymbol{f}}. \tag{60}$$

Using the conjugate-transpose identities for complex matrices, (60) can be rewritten in the compact complex-domain form

$$\Delta\widehat{\boldsymbol{v}}_{\mathrm{BLUE}} = \left(\Re\left\{\boldsymbol{H}^\dagger\boldsymbol{H}\right\}\right)^{-1}\Re\left\{\boldsymbol{H}^\dagger\Delta\boldsymbol{f}\right\}. \tag{61}$$

The locally linearized BLUE estimate is therefore $\widehat{\boldsymbol{v}}_{\mathrm{BLUE}} = \boldsymbol{v}_{\mathrm{ref}} + \Delta\widehat{\boldsymbol{v}}_{\mathrm{BLUE}}$.

Under a fixed reference point, $\boldsymbol{H}$ is deterministic. The error covariance matrix of the locally linearized BLUE is

$$\begin{aligned}\boldsymbol{C}_{\mathrm{BLUE}} &= \mathrm{Var}\left(\Delta\widehat{\boldsymbol{v}}_{\mathrm{BLUE}}\right)\\ &= \left(\Re\left\{\boldsymbol{H}^\dagger\boldsymbol{H}\right\}\right)^{-1}\mathrm{Var}\left(\Re\left\{\boldsymbol{H}^\dagger\boldsymbol{w}\right\}\right)\left(\Re\left\{\boldsymbol{H}^\dagger\boldsymbol{H}\right\}\right)^{-1}.\end{aligned} \tag{62}$$

Since $\mathrm{Var}(\Re\{\boldsymbol{H}^\dagger\boldsymbol{w}\}) = \sigma_w^2\Re\{\boldsymbol{H}^\dagger\boldsymbol{H}\}/2$, (62) simplifies to

$$\boldsymbol{C}_{\mathrm{BLUE}} = \frac{\sigma_w^2}{2}\left(\Re\left\{\boldsymbol{H}^\dagger\boldsymbol{H}\right\}\right)^{-1}. \tag{63}$$

Equation (63) highlights two key facts. First, if $\boldsymbol{v}_{\mathrm{ref}}$ approaches the true parameter $\boldsymbol{v}$, then $\boldsymbol{H}$ approaches the true Jacobian and $\boldsymbol{C}_{\mathrm{BLUE}}$ approaches the CRB. Second, when $\boldsymbol{v}_{\mathrm{ref}} \neq \boldsymbol{v}$, the neglected higher order term in (58) introduces linearization loss, so the practical estimator is generally suboptimal

$$\boldsymbol{C}_{\mathrm{BLUE}} = \boldsymbol{J}^{-1}\left(\boldsymbol{v}_{\mathrm{ref}}\right)$$

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$$\boldsymbol{J}(\boldsymbol{v}_{\mathrm{ref}}) = \frac{2}{\sigma_w^2}\Re\left\{\boldsymbol{H}^{\dagger}\boldsymbol{H}\right\}. \tag{64}$$

Two cases must be distinguished. When the reference vector v_{ref} is fixed independently of the current noisy observation, (63) gives the covariance of the locally linearized estimator conditional on that deterministic reference. When the reference is obtained from the same noisy observation, as in the FCA-initialized implementation considered in Section IV, both v_{ref} and $\mathbf{H}(\mathbf{v}_{\mathrm{ref}};\Xi)$ are data dependent. Equation (63) is therefore not the exact unconditional covariance of the resulting estimator. The latter case is referred to here as an FCA-initialized locally linearized correction implemented through the BLUE-form update, and its end-to-end performance is evaluated directly by paired Monte Carlo simulation. Equation (64) makes the fixed-reference local relation explicit. When the linearization is centered at the true parameter, $\mathbf{v}_{\mathrm{ref}} = \mathbf{v}$, the fixed-reference covariance expression reduces to the conditional CRB associated with the true Jacobian. For a mismatched reference, however, the neglected higher order terms introduce deterministic linearization error, so this covariance relation alone does not imply CRB-level total RMSE. The zero-mismatch case is therefore used only as an oracle theoretical consistency benchmark in the numerical study.

The approximation is intrinsically local and depends jointly on the reference mismatch $\|\boldsymbol{v}_{\mathrm{ref}} - \boldsymbol{v}\|_2$ and the receiver-noise level. A sufficiently small mismatch is required for the neglected higher order terms to remain insignificant. High SNR alone does not guarantee near-CRB behavior for a fixed nonzero mismatch, because the conditional CRB decreases with receiver-noise variance whereas the deterministic linearization error does not vanish at the same rate. The relevant local-validity behavior is therefore configuration dependent. In particular, the numerical region reported in Section IV is conditional on the prescribed scattering realization Ξ, receiving-array geometry, true wind vector, sampling configuration, and echo model adopted in this study; it should not be interpreted as a universal validity region for all MF radar operating conditions.

To expose how array geometry and coherent integration affect estimator performance, it is useful to expand the information term in (61). Partition the Jacobian matrix as $\boldsymbol{H} = [\boldsymbol{h}_x, \boldsymbol{h}_y]$, where $\boldsymbol{h}_x$ and $\boldsymbol{h}_y$ are the sensitivity vectors with respect to v_x and v_y, respectively.

The componentwise form of the BLUE update is provided in Appendix C.

The ideal BLUE would require a reference point arbitrarily close to the true wind vector and is therefore not directly realizable. In practice, $\boldsymbol{v}_{\mathrm{ref}}$ must be supplied by a coarse estimator or external prior information.

If the FCA estimate is used as the reference point, then the BLUE can be interpreted as an FCA estimate plus a Jacobian-based correction term. In that case

$$\Delta\widehat{\boldsymbol{v}} = \widehat{\boldsymbol{v}}_{\mathrm{BLUE}} - \widehat{\boldsymbol{v}}_{\mathrm{FCA}} = \left(\Re\left\{\boldsymbol{H}^{\dagger}\boldsymbol{H}\right\}\right)^{-1}\Re\left\{\boldsymbol{H}^{\dagger}\Delta\boldsymbol{f}\right\} \tag{65}$$

where $\Delta\boldsymbol{f} = \boldsymbol{f} - \boldsymbol{y}(\widehat{\boldsymbol{v}}_{\mathrm{FCA}})$. Expanding (65) componentwise yields

$$\begin{bmatrix} \hat{v}_{x,\mathrm{BLUE}} - \hat{v}_{x,\mathrm{FCA}} \\ \hat{v}_{y,\mathrm{BLUE}} - \hat{v}_{y,\mathrm{FCA}} \end{bmatrix} = \begin{bmatrix} E_{xx} & E_{xy} \\ E_{xy} & E_{yy} \end{bmatrix}^{-1} \begin{bmatrix} P_x \\ P_y \end{bmatrix}. \tag{66}$$

Equation (66) clarifies that the gap between BLUE and FCA is controlled jointly by the local curvature of the signal manifold and the residual mismatch at the reference point.

Equations (26), (63), and (66) provide complementary local interpretations of mechanisms that can contribute to the FCA–CRB performance margin. Correlation-parameter extraction and nonlinear mapping affect the FCA estimate, while the residual-driven BLUE correction characterizes the effect of local reference mismatch. These expressions do not constitute an exact or exhaustive decomposition of the entire FCA–CRB difference, particularly when the reference point lies outside the local linearization region. The numerical validity study in Section IV explicitly quantifies this limitation. Outside this regime, however, the higher order nonlinear terms omitted in (58) become nonnegligible, and the BLUE-based explanation should be viewed only as an approximation.

This observation also suggests a concrete direction for algorithm improvement. These expressions suggest that reducing correlation-parameter uncertainty, improving the conditioning of the FCA mapping, or reducing the local full-signal residual can potentially reduce the excess estimation error relative to the conditional benchmark. Hence, (26) and (65) together provide the theoretical basis for improving FCA or designing alternative estimators that more closely approach the CRB.

One additional caution is necessary. Equation (66) explains the discrepancy between BLUE and FCA only after a reference point has been fixed. It does not establish a global performance ordering among nonlinear estimators. That ordering still depends on how accurately the reference point approximates the true wind vector and on whether the neglected second-order terms remain small.

IV. Simulation and Verification

A. Numerical Simulation Examples

This section validates the derived CRB via numerical simulations.

The following numerical values are assigned to the symbolic model parameters. The receiving array uses $N = 3$ channels with $\boldsymbol{d}_1 = [0, 115, 0]^{\dagger}$, $\boldsymbol{d}_2 = [-100, -57, 0]^{\dagger}$, and $\boldsymbol{d}_3 = [100, -57, 0]^{\dagger}$. For this triangular array, the stacked linear baseline matrix satisfies $\mathrm{rank}(\mathbf{B}) = 2$, and the corresponding quadratic baseline matrix satisfies $\mathrm{rank}(\mathbf{D}) = 3$. Hence, both (f, g) and (a, b, h) are identifiable for the numerical configuration used here. The scatterer support introduced in the echo model is set to $[x_{\min}, x_{\max}] = [-20, 20]$ km, $[y_{\min}, y_{\max}] = [-20, 20]$ km, and $[z_{\min}, z_{\max}] = [84, 96]$ km. The number of scatterers is fixed at $N_{\mathrm{sca}} = 2000$, and the initial positions are drawn uniformly as $r_{i,x} = \mathrm{rand}([x_{\min}, x_{\max}])$, $r_{i,y} = \mathrm{rand}([y_{\min}, y_{\max}])$, and $r_{i,z} = \mathrm{rand}([z_{\min}, z_{\max}])$. The background horizontal wind magnitude is set to 50 m/s. For the vertically pointing beam model in (81), the beam parameters are specified as $\theta_a = 0$ and $\theta_t = 15°$, so that $\sin\theta_t \approx 0.26$ in the generated data. The spatial geometry of the receiving array and the scatterers is illustrated in Fig. 2.

The red dots represent the three spatially separated MF radar receiving antennas. Figs. 3 and 4 show the received complex time series (real and imaginary parts) and amplitudes for these three channels. The blue solid, red dashed, and black dotted

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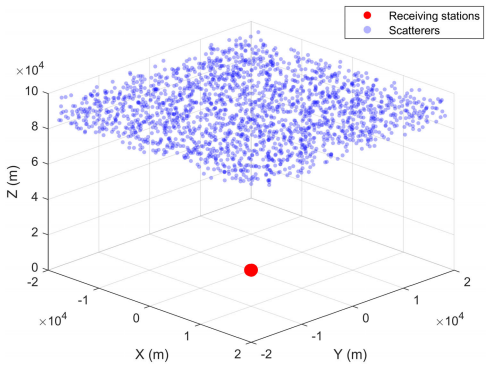


Fig. 2. Spatial distribution of receiving stations and scatterers.

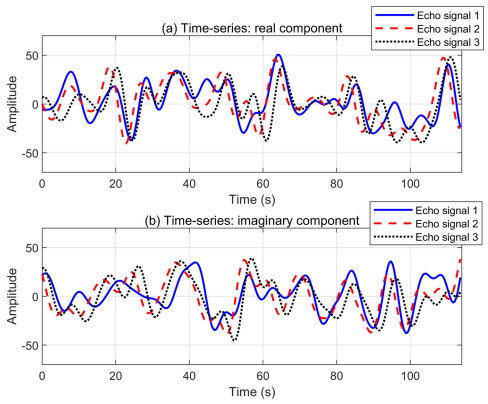


Fig. 3. Simulated complex time series (real and imaginary parts). (a) Real components of the simulated complex echo signals. (b) Imaginary components of the simulated complex echo signals.

curves correspond to Antennas 1–3, respectively. The three time series exhibit clear relative delays in the complex plane, consistent with the expected FCA echo behavior.

To further investigate the ground diffraction characteristics, a rectangular observation area covering 600 m in both the zonal and meridional directions is defined. A regular grid of 300×300 reception points is then placed in this area with a spacing of 2 m. This produces 90 000 reception points in total. Echo amplitudes are extracted at all grid points, and contour maps are generated at 2-s intervals. As shown in Fig. 5, the contours migrate clearly toward the upper right. This trend agrees with the preset background wind direction. The elliptical contour shape also supports the theoretical assumptions of FCA.

For the simulation study, the receiver-noise variance is set equal across channels, i.e., $\sigma_w^2 = \sigma^2$. The SNR is defined as follows:

$$\mathrm{SNR} = 10\log_{10}\frac{\sum_{n=1}^{N}\sum_{k=1}^{K}|y_n[k]|^2}{KN\sigma^2}\ \mathrm{dB}. \tag{67}$$

Based on (54) and (55), the standard-deviation lower bounds are

$$\mathrm{STD}_{v_x} = \sqrt{\mathrm{CRB}_{v_x}} \tag{68}$$

$$\mathrm{STD}_{v_y} = \sqrt{\mathrm{CRB}_{v_y}}. \tag{69}$$

These expressions give the standard-deviation lower bounds of v_x and v_y. Because the conventional CRB is a covariance

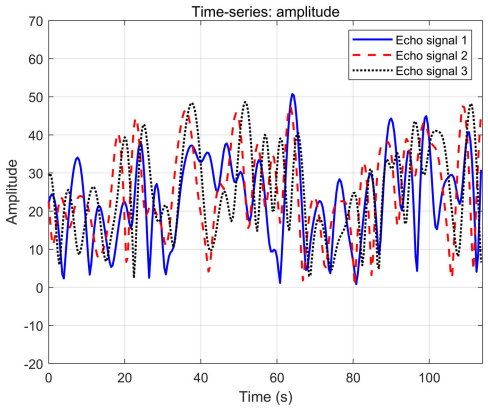


Fig. 4. Simulated complex time series (amplitude).

bound for unbiased or locally unbiased estimation, Fig. 6 shows the comparison of the estimator standard deviations with the square-root CRB rather than comparing RMSE directly with the CRB. Bias and total error are examined separately below.

The numerical conditional MLE is evaluated using 5000 Monte Carlo trials. Fig. 6 shows the comparison of the its empirical standard deviation with the square-root conditional CRB and therefore evaluates estimator dispersion rather than total mse. To examine the unbiasedness assumption independently, Fig. 7(a) reports the signed Monte Carlo bias, while Fig. 7(b) reports the corresponding RMSE. The CRB is not plotted as an RMSE lower bound in Fig. 7(b) because the conventional CRB used here bounds the covariance of unbiased or locally unbiased estimators and should not be interpreted as a general RMSE lower bound for biased FCA estimates. The MLE bias remains very small over the investigated SNR values, with a maximum absolute bias of approximately 0.055 m/s for either wind component. Its squared-bias contribution remains below approximately 0.5 % of the mse, so that the MLE RMSE is practically indistinguishable from its standard deviation. The small finite-sample bias supports treating the conditional MLE as approximately locally unbiased over the investigated conditional simulation setting; it does not by itself establish asymptotic unbiasedness. The approach of the empirical MLE standard deviation to the conditional CRB scale at high SNR is consistent with the asymptotic-efficiency behavior expected under the adopted model.

One realization of the $N_{\mathrm{sca}} = 2000$ initial scatterer positions is generated and then held fixed throughout the Monte Carlo experiment. For each SNR, independent AWGN realizations are regenerated across the 5000 trials, while the scattering realization remains unchanged. The numerical conditional MLE uses the same fixed realization that is used to generate the noiseless observations. Accordingly, the reported MLE statistics and CRB are conditional on this particular scattering realization. Holding this scattering realization fixed across the Monte Carlo trials defines a conditional repeated-noise experiment. It should not be interpreted as an assumption that the physical atmosphere remains unchanged over multiple observation windows. Within each simulated observation sequence, the scatterers continue to advect according to (27).

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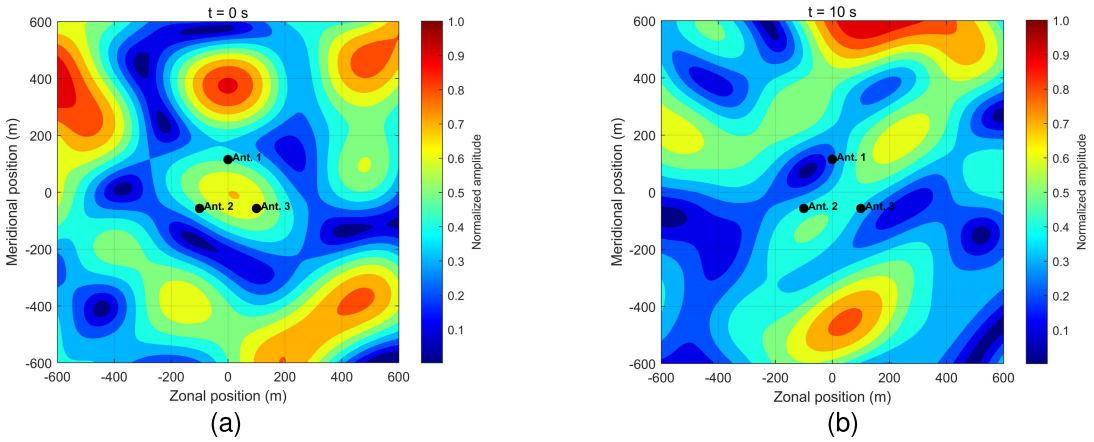


Fig. 5. Evolution of the ground diffraction pattern generated by the model. (a) Ground diffraction pattern at $t = 0$ s. (b) Ground diffraction pattern at $t = 10$ s.

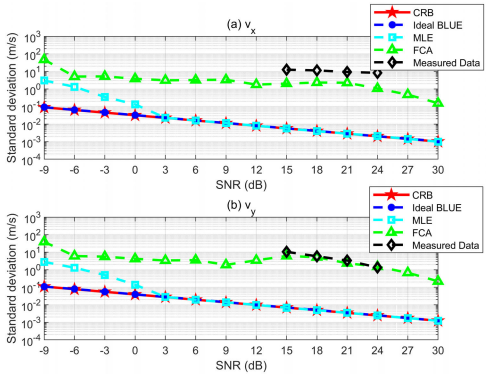


Fig. 6. Componentwise estimator standard deviations and corresponding square-root conditional CRB scales under the prescribed scattering realization. (a) Standard deviations and corresponding conditional CRB scales for the zonal wind component (v_x). (b) Standard deviations and corresponding conditional CRB scales for the meridional wind component (v_y).

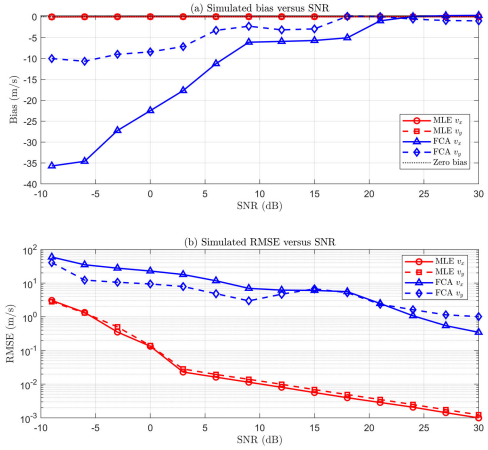


Fig. 7. Monte Carlo signed bias and RMSE of the conditional full-signal MLE and conventional FCA from 5000 receiver-noise realizations under the same prescribed scattering realization. (a) Signed estimation bias versus SNR. (b) Estimation RMSE versus SNR.

In view of (45)–(47), the MLE in Fig. 6 plays two roles. It numerically approximates the optimum nonlinear estimator under the adopted signal model. It also provides empirical support that the derived CRB is tight in the asymptotic regime. The convergence of the MLE curve toward the CRB should therefore be interpreted as evidence for the correctness of the CRB derivation, not as a claim that the MLE admits a closed-form solution. The BLUE curve in Fig. 6 corresponds to the ideal zero-mismatch case $v_{\mathrm{ref}} = v$. It is included as a local theoretical benchmark and is not a directly realizable estimator because it requires knowledge of the true wind vector.

The FCA method is then applied to the simulated data using the same fixed correlation-processing configuration at all investigated SNR values. Specifically, an eighth-order polynomial fitting procedure is used over the lag interval $l = -10, \ldots, 10$, corresponding to 21 lag samples. With $T_s = 0.4$ s, this interval corresponds to -4 to 4 s. The polynomial order and lag interval are held fixed throughout the numerical study and are not reoptimized for individual SNR values. The resulting correlation quantities are then used to estimate the FCA intermediate parameters and retrieve the horizontal wind components. For the three-receiver configuration, correlation quantities from the three baselines (1, 2), (2, 3), and (3, 1) are used. The intermediate parameters f and g, followed by a, b, and h, are obtained using least-squares pseudoinverse solutions before the standard FCA wind-velocity expressions are evaluated.

The FCA standard-deviation curves in Fig. 6 lie above the corresponding square-root conditional CRB scales over much of the investigated SNR range. However, the additional bias results in Fig. 7(a) show that conventional FCA cannot generally be treated as locally unbiased under the present processing settings. In particular, substantial systematic bias occurs over several SNR values. For example, at 0 dB, the biases of v_x and v_y are approximately -22.48 and -8.42 m/s, respectively, while their standard deviations are approximately 3.85 and 4.22 m/s. Consequently, the corresponding RMSE values increase to approximately 22.81 and 9.42 m/s, as shown in Fig. 7(b).

Fig. 6 places the empirical FCA dispersion relative to the conditional full-signal benchmark for the prescribed simulation setting. Although the FCA observations, conditional MLE, and conditional CRB are based on data generated under the same scattering realization and receiver-noise model, the compared methods do not operate with identical auxiliary information. The conditional MLE and CRB explicitly use the prescribed realization Ξ when constructing the full-signal template, whereas conventional FCA is not supplied with that microscopic scattering realization. Therefore, the

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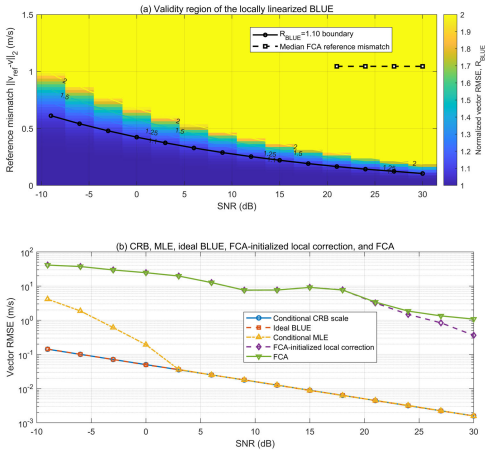


Fig. 8. Numerical local-validity analysis of the locally linearized full-signal correction under the adopted fixed scattering realization. (a) Normalized vector RMSE as a function of SNR and reference-wind mismatch, showing the 10%-near-CRB boundary. (b) Comparison of vector RMSE versus SNR for the conditional CRB scale, ideal BLUE, conditional MLE, FCA-initialized local correction, and conventional FCA.

separation between the FCA dispersion and the conditional CRB scale should not be interpreted as pure information loss caused by FCA processing. It may reflect both the effects of correlation-based processing and the additional scattering-realization knowledge assumed by the conditional benchmark. The present experiment does not isolate or quantify these two contributions. The covariance-propagation relation in (26) identifies important estimator-level error mechanisms within FCA, but it does not provide a decomposition of the complete FCA–CRB separation. The FCA curve in Fig. 6 should therefore be interpreted as the dispersion component of the estimation error rather than as the total FCA retrieval error. The RMSE in Fig. 7(b) accounts for both the variance and bias components. These results also show that the local covariance propagation in (26) characterizes an important variance mechanism of FCA, but does not by itself represent the complete mse when systematic bias is significant.

To characterize the local validity behavior of the linearized correction, the SNR is varied over $-9, -6, \ldots, 30$ dB, and the reference mismatch $\rho = \|\mathbf{v}_{\text{ref}} - \mathbf{v}\|_2$ is varied from 0 to 1.5 m/s with a step of 0.025 m/s. For each mismatch magnitude, eight uniformly spaced directions $\theta = 0°, 45°, \ldots, 315°$ are considered, with $\mathbf{v}_{\text{ref}} = \mathbf{v} + \rho[\cos\theta, \sin\theta]^{\dagger}$. The conditional signal template and Jacobian are recomputed at every reference point. For a fixed reference, the direction-dependent vector mse is evaluated directly from the exact bias of the locally linearized update under the adopted nonlinear echo model and its fixed-reference covariance: $\text{mse}_{\text{vec}} = \|\mathbf{b}_{\text{ref}}\|_2^2 + \text{tr}(\mathbf{C}_{\text{ref}})$. Consequently, no receiver-noise Monte Carlo simulation is required for Fig. 8(a). For each (ρ, SNR) pair, the vector mse is averaged over the eight mismatch directions, and the normalized vector RMSE is defined as $R_{\text{BLUE}} = \sqrt{\overline{\text{mse}}_{\text{dir}}} / \sqrt{\text{tr}[\mathbf{C}_{\text{CRB}}(\mathbf{v} \mid \Xi)]}$. The contour $R_{\text{BLUE}} = 1.10$ is used only as an operational criterion for identifying a direction-averaged RMSE within 10% of the conditional CRB scale. The reported boundary is obtained by interpolation between adjacent mismatch-grid points and does not define a universal BLUE-validity theorem.

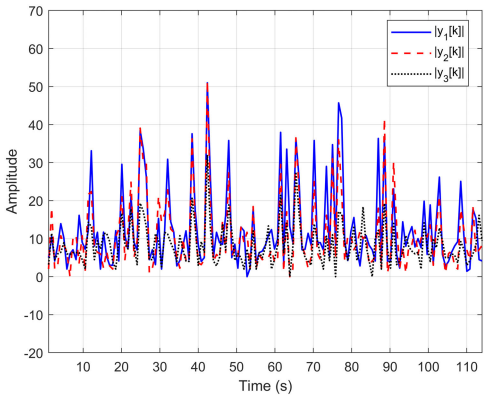


Fig. 9. Actual received signals from the three channels of the MF radar.

TABLE II

MAIN EXPERIMENTAL AND PROCESSING PARAMETERS FOR THE MEASURED-DATA ANALYSIS

Parameter	Value / setting
Observation date	July 24, 2023
MF radar location	Yunnan, China
MF radar operating frequency	2.138 MHz
Number of receiving channels	3
Receiving-array geometry	Equilateral triangular array
Sampling interval	0.4 s
Analyzed altitude interval	84–96 km
Representative displayed altitude	88 km
Representative sequence duration	114 s
Retained SNR range	15–24 dB
FCA polynomial order	8
FCA lag interval	$l = -10, \ldots, 10$
Number of FCA lag samples	21
Corresponding lag-time interval	-4 to 4 s
Meteor radar temporal resolution	2 h
Altitude resolution for wind comparison	2 km

In the investigated simulation configuration, the adopted 10%-near-CRB criterion yields reference-mismatch thresholds of approximately 0.424 m/s at 0 dB and 0.105 m/s at 30 dB. These values characterize the reported mismatch experiment and operational criterion rather than universal requirements for MF radar wind retrieval. The trend shows that increasing SNR alone does not ensure near-bound local correction: as the noise-limited conditional CRB decreases, the reference must become more accurate to keep the deterministic linearization error within the same relative distance of the bound.

The FCA-initialized experiment further distinguishes useful estimator improvement from attainment of the conditional information benchmark. At 30 dB, the FCA-initialized correction reduces vector RMSE by approximately 66.5% relative to FCA; yet, its RMSE remains approximately 0.362 m/s, compared with the conditional CRB scale $\sqrt{\text{tr}(\mathbf{C}_{\text{CRB}})} \approx 1.59 \times 10^{-3}$ m/s. Thus, a substantial reduction in FCA error does not imply that the conditional bound has been approached. The correction remains dependent on the prescribed scattering template Ξ, so this improvement is not presented as an operationally attainable gain under unknown atmospheric scattering.

B. Measured-Data Illustration and External-Reference Consistency Analysis

The practical value of any radar signal processing algorithm derived from simulation depends on whether the

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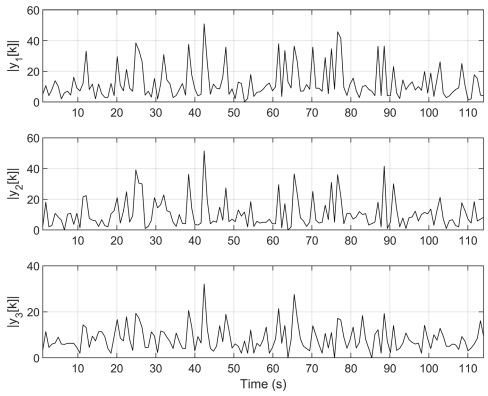


Fig. 10. Observed complex sequences from the three channels within the 88-km altitude window over a duration of 114 s.

physical model represents the sounding environment with sufficient fidelity. Figs. 3 and 4 have already shown the simulated received sequences. The corresponding measured echo sequences are presented in Figs. 9 and 10. The principal experimental and processing parameters used in the measured-data analysis are summarized in Table II. The MF radar observations were acquired in Yunnan, China, on July 24, 2023 using a three-channel system operating at 2.138 MHz. The receiving antennas form an equilateral triangular array. The complex received sequences are sampled at $T_s = 0.4$ s, and the altitude interval considered in this study is 84–96 km. The representative measured sequence shown below is centered at 88 km and spans 114 s. The measured signals exhibit pronounced random amplitude fluctuations and macroscopic fading behavior. The simulated signals show qualitatively similar fading patterns and temporal fluctuation characteristics. This resemblance provides a qualitative plausibility check that the adopted echo model reproduces selected observable features of the measured MF radar echoes. However, such qualitative agreement does not constitute validation of the complete statistical echo model or of all assumptions entering the conditional signal formulation. Accordingly, the performance bounds and estimator comparisons reported in this work should be interpreted as model-based results under the adopted echo model rather than as universally validated limits for measured MF radar data. In particular, the measured observations are not assumed to correspond to the same prescribed scattering realization Ξ used in the conditional simulation. The simulated–measured comparison is therefore used only to assess qualitative similarity in selected observable echo characteristics and is not a validation of the particular conditional CRB for the measured atmosphere.

For the cross-instrument comparison, the meteor radar wind product, which has a temporal resolution of 2 h, is used to define the common temporal grid. For each meteor radar 2-h interval, all MF radar results falling within the corresponding 2-h interval are averaged to obtain the MF radar value used for comparison. The comparison is therefore performed on a common 2-h temporal basis rather than between the native 0.4-s MF radar sampling and the 2-h meteor radar product. Both radar products are compared at a 2-km altitude resolution. The corresponding altitude levels are therefore matched directly without additional altitude interpolation. The common altitude interval of 84–96 km is selected because both radar systems provide clear usable observations throughout this shared height range. For the SNR-dependent measured-data statistics, the measured SNR values are first rounded to the nearest integer dB. Measurements assigned to the same integer-SNR value are then averaged to obtain the reported SNR-dependent quantities.

Measured observations are subsequently used for external radar intercomparison and for illustrating reference-assisted local correction behavior on real MF radar data; they are not used as independent validation of the conditional CRB, the complete echo model, or standalone estimator accuracy. Observations were obtained from the meteor radar and MF radar stationed in Yunnan and provided by CETC 22 on July 24, 2023. Measured MF radar records from the 84–96 km altitude interval with SNR values between 15 and 24 dB are considered in the radar intercomparison. Since the true atmospheric wind speed is unknown, the meteor radar measurements are used as an external reference for the measured FCA comparison. The measured-data result is shown by the black curve in Fig. 6. Its standard deviation varies with altitude because the actual atmospheric wind field and scattering environment are variable. Importantly, the measured observations are not assumed to share the prescribed scattering realization Ξ used to calculate the simulated conditional CRB and MLE curves. Therefore, the separation between the measured curve and the conditional CRB should be interpreted only as an external empirical comparison with the model-based benchmark, rather than as a direct measurement of the distance between measured FCA and its own conditional performance bound. In the displayed comparison, the measured FCA dispersion is numerically higher than the conditional simulation benchmark. However, because the measured observations do not share the prescribed scattering realization Ξ used in the simulation and the true atmospheric wind is unknown, this comparison does not quantify the distance between the measured FCA estimator and its own conditional CRB. It is therefore interpreted only as an external empirical radar intercomparison.

The locally linearized BLUE should therefore be interpreted as a correction mechanism rather than a universally superior replacement for FCA. The ideal version of BLUE requires the true wind vector to compute the Jacobian matrix and is not physically realizable. Two BLUE configurations are distinguished in the simulation study. The ideal BLUE in Fig. 6 uses $v_{\mathrm{ref}} = v$ and therefore serves only as a zero-mismatch theoretical benchmark. The FCA-initialized BLUE in Fig. 8 obtains its reference wind from the FCA estimate of the same noisy observation, so no external wind reference is required. However, evaluation of $\mathbf{y}(\hat{\mathbf{v}}_{\mathrm{FCA}}; \Xi)$ and the corresponding Jacobian still requires the prescribed scattering realization Ξ. Thus, the FCA-initialized experiment represents a local correction under known-template conditions rather than a self-contained solution to the unknown-scattering problem. Neither configuration is interpreted as a globally optimal operational estimator. The measured-data experiment considered next represents a third configuration: an externally assisted BLUE, for which the reference wind is supplied by the meteor radar. Because this

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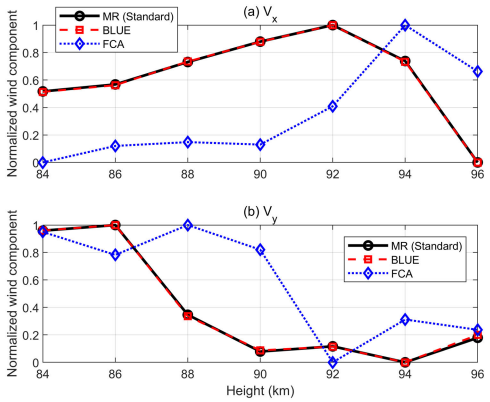


Fig. 11. Measured-data illustration of the externally assisted local correction using meteor radar wind as the prescribed external reference. (The plotted quantities are independently min–max normalized wind-component profiles over the displayed altitude grid. The comparison illustrates reference-conditioned profile behavior and does not constitute independent validation of standalone estimator accuracy). (a) Independently min–max normalized zonal wind component (v_x) profiles versus altitude. (b) Independently min–max normalized meridional wind component (v_y) profiles versus altitude.

external wind enters the estimator itself, subsequent agreement with the meteor radar cannot be used as an independent accuracy validation. The purpose of the experiment is instead to illustrate the behavior of the local correction on real MF radar observations when an external reference is available. The ideal BLUE in Fig. 6 uses a zero-mismatch reference and therefore serves only as a local theoretical benchmark under the prescribed scattering template. Reference-mismatch effects and the FCA-initialized correction are evaluated separately in Fig. 8, conditional on the same prescribed Ξ. Neither comparison decomposes the complete FCA–CRB separation.

The BLUE curve in Fig. 6 should therefore be interpreted as an ideal zero-mismatch local benchmark rather than as a directly realizable estimator. The effect of a nonzero reference mismatch, including the FCA-initialized case, is examined separately in Fig. 8. As the mismatch increases, the neglected higher order terms of the local linearization become increasingly important, so the locally linearized correction can remain substantially above the conditional CRB even when it improves upon the original FCA estimate.

To illustrate the behavior of the locally linearized correction on real MF radar observations when an external reference is available, collocated meteor radar and MF radar measurements from July 24, 2023 are considered. The meteor radar wind is supplied to BLUE as the external reference v_{ref}. Consequently, this experiment is externally assisted by construction and is not intended as an independent validation of BLUE accuracy. Its purpose is to examine whether the measured-data correction behaves consistently with the reference-dependent mechanism identified in the theoretical and simulation analyses. For comparison with the meteor radar, the same 84–96-km altitude interval is considered with a 2-km altitude resolution. The meteor radar wind product has a temporal resolution of 2 h. The two products are compared on the common 2-km altitude grid over 84–96 km. Because the measurement principles and temporal resolutions differ, the comparison is used only for an altitude-profile illustration and external intercomparison; pointwise differences are not interpreted as direct measurements of the unknown true-wind error. For the profile illustration in Fig. 11, the altitude-dependent wind-component profiles obtained from conventional FCA, the externally assisted local correction, and meteor radar are independently min–max normalized over the displayed altitude grid. For a profile $v(z_j)$, the normalization is $\tilde{v}(z_j) = [v(z_j) - \min_j v(z_j)]/[\max_j v(z_j) - \min_j v(z_j)]$. The same normalization is applied separately to each method and to each horizontal wind component. Fig. 11 therefore shows the comparison of normalized altitude-profile shapes rather than correlation coefficients or absolute wind-speed errors.

Fig. 11 shows the comparison of the independently min–max normalized altitude profiles of the conventional FCA estimate, the externally assisted local correction, and the meteor radar wind used as the external reference. After this external reference is supplied to the correction, the resulting normalized MF radar profile is more similar in its altitude-dependent variation to that same reference than the original FCA profile. This behavior is consistent with the expected response of a reference-assisted local correction. It illustrates that the Jacobian-based full-signal update can modify the FCA estimate in a direction consistent with an externally supplied wind reference. Because the meteor radar wind has already entered the correction through v_{ref}, however, the subsequent profile agreement is reference-conditioned by construction. Accordingly, Fig. 11 should be regarded as an illustration of reference-conditioned behavior under external assistance, rather than as independent evidence of estimator accuracy or as proof that the corrected estimate is closer to the unknown true atmospheric wind than conventional FCA. Independent validation of such an accuracy claim would require an additional wind reference that is not used in forming the correction.

V. Conclusion

The principal contribution of this work is an analytical Fisher-information/CRB benchmark for horizontal wind estimation under the adopted MF radar echo model and a fixed and known scattering realization Ξ. It complements earlier estimator-specific FCA/spaced-antenna error analyses under an explicitly stronger scattering-information condition. Conditional MLE simulations provide numerical evidence of high-SNR tightness within this model, rather than a general proof of attainability for measured atmospheric winds.

The local correction analysis quantifies the reference accuracy needed to approach the conditional benchmark in the investigated configuration. Under the 10%-near-CRB criterion, the admissible reference mismatch decreases from approximately 0.424 m/s at 0 dB to 0.105 m/s at 30 dB. At 30 dB, FCA initialization followed by one full-signal correction reduces vector RMSE by approximately 66.5%, yet, the resulting 0.362 m/s remains well above the conditional CRB scale of approximately 1.59×10^{-3} m/s. Thus, higher SNR and useful local improvement do not, by themselves, imply attainment of the conditional bound. The signal template and Jacobian in this correction continue to require the prescribed scattering information.

Conventional FCA is compared with this idealized benchmark using observations generated under the same prescribed

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realization. Because its auxiliary information differs from that of the conditional benchmark, the reported FCA–CRB separation does not isolate FCA processing loss or quantify a gain available from FCA improvements alone. The local covariance analysis describes contributing estimator-error mechanisms, while bias and RMSE are reported separately from dispersion. Measured observations provide qualitative checks of selected echo characteristics and illustrate correction conditioned on an external wind reference; they do not independently validate the conditional CRB or the accuracy of the externally assisted estimator. Extending the benchmark to unknown or uncertain scattering information remains a separate statistical problem.

Appendix A
Analytical Details of FCA Correlation-Parameter Extraction

For the local analytical FCA error formulation introduced in Section II, the logarithm of the sampled cross-correlation magnitude is approximated by a quadratic model in a symmetric neighborhood of the coarse peak. This approximation is introduced for analytical tractability and is distinct from the eighth-order polynomial fitting procedure used in the numerical FCA implementation of Section IV.

Define the local data vector and design matrix as follows:

$$\boldsymbol{z}_{nn'} = \left[\ln\left|\rho_{nn'}\left(\tau_{\hat{l}_{nn'}-M}\right)\right|, \ldots, \ln\left|\rho_{nn'}\left(\tau_{\hat{l}_{nn'}+M}\right)\right|\right]^{\dagger} \tag{70}$$

$$\boldsymbol{X} = \left[\mathbf{1}, \boldsymbol{\delta}, \boldsymbol{\delta}^{\circ 2}\right], \quad \boldsymbol{\delta} = [-MT_{\mathrm{s}}, \ldots, MT_{\mathrm{s}}]^{\dagger} \tag{71}$$

where $(\cdot)^{\circ 2}$ denotes the elementwise square. The least-squares estimate of the quadratic coefficients is

$$\hat{\boldsymbol{c}}_{nn'} = \left(\boldsymbol{X}^{\dagger}\boldsymbol{X}\right)^{-1}\boldsymbol{X}^{\dagger}\boldsymbol{z}_{nn'} \tag{72}$$

$$\hat{\boldsymbol{c}}_{nn'} = \left[\hat{c}_{0,nn'}, \hat{c}_{1,nn'}, \hat{c}_{2,nn'}\right]^{\dagger}. \tag{73}$$

The interpolated peak lag and the decorrelation time are then estimated by

$$\hat{\tau}'_{nn'} = \tau_{\hat{l}_{nn'}} - \frac{\hat{c}_{1,nn'}}{2\hat{c}_{2,nn'}} \tag{74}$$

$$\left(\hat{\tau}''_{nn'}\right)^2 = -\frac{1}{\hat{c}_{2,nn'}} \tag{75}$$

where $\hat{c}_{2,nn'} < 0$ is required for a valid local maximum. Stacking the peak-lag and decorrelation-time estimates over all antenna pairs gives

$$\hat{\boldsymbol{\tau}}' = \left[\hat{\tau}'_{12}, \ldots, \hat{\tau}'_{nn'}, \ldots, \hat{\tau}'_{N1}\right]^{\dagger} \tag{76}$$

$$\left(\hat{\boldsymbol{\tau}}''\right)^2 = \left[\left(\hat{\tau}''_{12}\right)^2, \ldots, \left(\hat{\tau}''_{nn'}\right)^2, \ldots, \left(\hat{\tau}''_{N1}\right)^2\right]^{\dagger}. \tag{77}$$

Appendix B
Conditional Echo-Model Derivatives and Fisher-Information Evaluation

A. Echo Signal Model

Starting from the factorized amplitude and propagation-phase model introduced in Section III-A, this appendix gives the intermediate expressions leading to the compact conditional echo vector used for MLE and Fisher-information evaluation.

The amplitude of the received signal can be expanded as follows:

$$\alpha_n\left(\boldsymbol{r}_i[k]\right) = \sqrt{p_i P\left(\boldsymbol{r}_i[k]\right) S\left(\boldsymbol{r}_i[k]\right) R\left(\boldsymbol{r}_i[k]\right)} \tag{78}$$

where p_i denotes the effective scattering coefficient of the ith scatterer, $P(\boldsymbol{r}_i[k])$ represents the combined transmit-receive pattern, $S(\boldsymbol{r}_i[k])$ indicates the effective polarization diagram of the ith scatterer, and $R(\boldsymbol{r}_i[k])$ denotes the range-gate function. In the adopted simulation model, all scatterers are assigned the same prescribed scattering coefficient. This common coefficient is absorbed into the overall amplitude normalization and is used consistently in the noiseless echo template and its wind derivatives. Although the scattering coefficient is common and fixed, the contribution of each scatterer remains position-dependent through the amplitude and propagation-phase terms evaluated at $\mathbf{r}_i[k]$. The term $P(\boldsymbol{r}_i[k])$ is modeled as the product of the transmitting pattern $T(\boldsymbol{r}_i[k])$ and the receiving pattern $D(\boldsymbol{r}_i[k])$. Specifically,

$$P\left(\boldsymbol{r}_i[k]\right) = T\left(\boldsymbol{r}_i[k]\right) D\left(\boldsymbol{r}_i[k]\right) \tag{79}$$

where $T(\boldsymbol{r}_i[k])$ can be expanded as follows:

$$T\left(\boldsymbol{r}_i[k]\right) = \exp\left\{-\frac{\left|\boldsymbol{r}_i[k]\right|^2 - \left(\boldsymbol{r}_i[k]\cdot\boldsymbol{r}_a\right)^2}{\left(\left|\boldsymbol{r}_i[k]\right|\sin\theta_t\right)^2}\right\}. \tag{80}$$

In (80), θ_t represents the beamwidth. Assuming the transmitting beam has a tilt angle of θ_a with respect to the zenith (positive z-axis) and an azimuth angle of ϕ_a, then $\boldsymbol{r}_a = [\sin\phi_a\sin\theta_a, \cos\phi_a\sin\theta_a, \cos\theta_a]^{\dagger}$ denotes the unit vector of the beam direction. To simplify the theoretical analysis, the beam is further assumed to point vertically toward the zenith, i.e., $\theta_a = 0$ and $\boldsymbol{r}_a = [0, 0, 1]^{\dagger}$, while the beamwidth parameter θ_t is kept symbolic at this stage. Then,

$$T\left(\boldsymbol{r}_i[k]\right) \approx \exp\left\{-\frac{\left|\boldsymbol{r}_i[k]\right|^2 - \left(\boldsymbol{r}_{i,z}[k]\right)^2}{\left(\left|\boldsymbol{r}_i[k]\right|\sin\theta_t\right)^2}\right\}. \tag{81}$$

Under a uniform horizontal wind field, the vertical coordinate $r_{i,z}[k]$ is constant, so $T(\boldsymbol{r}_i[k]) \approx \alpha_1$, where α_1 is independent of the wind parameters to be estimated. Since the receiving-antenna directivity is typically broader than the scatterer aspect sensitivity, it does not introduce significant angular discrimination, and $D(\boldsymbol{r}_i[k]) = 1$ is adopted. The term $S(\boldsymbol{r}_i[k])$ characterizes the aspect sensitivity of the scatterers and can be written as follows:

$$S\left(\boldsymbol{r}_i[k]\right) = \exp\left\{-\frac{\left|\boldsymbol{r}_i[k]\right|^2 - \left(\boldsymbol{r}_i[k]\cdot\boldsymbol{r}_{si}\right)^2}{\left(\left|\boldsymbol{r}_i[k]\right|\sin\theta_s\right)^2}\right\}. \tag{82}$$

In (82), θ_s denotes the e^{-1} angular sensitivity width and $\boldsymbol{r}_{si}$ is the corresponding unit direction vector. For the vertically pointing configuration considered here, $\boldsymbol{r}_{si} = [0, 0, 1]^{\dagger}$ is adopted, and the aspect-sensitivity term is approximated by a constant α_2 that is independent of the wind parameters. The range-gate term $R(\boldsymbol{r}_i[k])$ is

$$R\left(\boldsymbol{r}_i[k]\right) = \exp\left\{-\frac{\left(\left|\boldsymbol{r}_i[k]\right| - r_0\right)^2}{r_e^2}\right\} \tag{83}$$

where r_0 denotes the effective center height of the range gate and r_e is the e^{-1} width of the range gate. After absorbing

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all factors that are independent of the wind parameters into a single scale factor, the amplitude can be simplified to

$$\alpha_n\left(\boldsymbol{r}_i[k]\right) \propto \left\{\exp\left\{-|\boldsymbol{r}_i[k]|^2\right\}\right\}^{1/2}. \tag{84}$$

The phase of the received signal can be expanded as follows:

$$\phi_n\left(\boldsymbol{r}_i[k]\right) = 2\pi\left(\frac{|\boldsymbol{r}_i[k]-\boldsymbol{t}|^2+|\boldsymbol{r}_i[k]-\boldsymbol{d}_n|^2}{\lambda}\right) \tag{85}$$

where $\boldsymbol{t}$ is the equivalent geometric center of the transmitting antenna array, $\boldsymbol{d}_n$ is the location of the nth receiving antenna, and λ is the wavelength of the transmitted signal. Without loss of generality, the coordinate origin is chosen at the equivalent geometric center of the transmitting antenna array, i.e., $\boldsymbol{t} = [0,0,0]^\dagger$. Consequently, the phase expression simplifies to

$$\phi_n\left(\boldsymbol{r}_i[k]\right) \propto \left(|\boldsymbol{r}_i[k]|^2+|\boldsymbol{r}_i[k]-\boldsymbol{d}_n|^2\right). \tag{86}$$

B. Conditional MLE and CRB

The partial derivative of the noiseless signal vector $\boldsymbol{y}$ with respect to the zonal velocity component v_x is

$$\frac{\partial \boldsymbol{y}}{\partial v_x} = \begin{bmatrix} \frac{\partial y_1[1]}{\partial v_x} \\ \vdots \\ \frac{\partial y_n[k]}{\partial v_x} \\ \vdots \end{bmatrix} \tag{87}$$

$$\frac{\partial y_n[k]}{\partial v_x} = \sum_{i=1}^{N_{\text{sca}}} \gamma_i^{(x)} \exp\left\{-\left(\frac{1}{2}+j\right)A - jB\right\} \tag{88}$$

where

$$\begin{aligned}\gamma_i^{(x)} = &-\left(\frac{1}{2}+j\right)2kT_{\text{s}}\left(r_{i,x}+v_x kT_{\text{s}}\right) \\ &- j2kT_{\text{s}}\left(r_{i,x}+v_x kT_{\text{s}}-x_n^{\text{r}}\right).\end{aligned} \tag{89}$$

By symmetry, the partial derivative with respect to the meridional velocity component v_y is given by

$$\frac{\partial y_n[k]}{\partial v_y} = \sum_{i=1}^{N_{\text{sca}}} \gamma_i^{(y)} \exp\left\{-\left(\frac{1}{2}+j\right)A - jB\right\} \tag{90}$$

where

$$\begin{aligned}\gamma_i^{(y)} = &-\left(\frac{1}{2}+j\right)2kT_{\text{s}}\left(r_{i,y}+v_y kT_{\text{s}}\right) \\ &- j2kT_{\text{s}}\left(r_{i,y}+v_y kT_{\text{s}}-y_n^{\text{r}}\right).\end{aligned} \tag{91}$$

Substituting (87)–(91) into (48) gives the FIM entries

$$[\boldsymbol{J}(\boldsymbol{v})]_{11} = \frac{2}{\sigma_w^2}\sum_{n=1}^{N}\sum_{k=1}^{K}\left|\frac{\partial y_n[k]}{\partial v_x}\right|^2 \tag{92}$$

$$[\boldsymbol{J}(\boldsymbol{v})]_{22} = \frac{2}{\sigma_w^2}\sum_{n=1}^{N}\sum_{k=1}^{K}\left|\frac{\partial y_n[k]}{\partial v_y}\right|^2 \tag{93}$$

$$[\boldsymbol{J}(\boldsymbol{v})]_{12} = [\boldsymbol{J}(\boldsymbol{v})]_{21} = \frac{2}{\sigma_w^2}\Re\left[\sum_{n=1}^{N}\sum_{k=1}^{K}\left(\frac{\partial y_n[k]}{\partial v_x}\right)^*\frac{\partial y_n[k]}{\partial v_y}\right]. \tag{94}$$

Appendix C
Derivation of the Locally Linearized BLUE

For completeness, the compact 2×2 BLUE relation can be expanded into a componentwise form. Define

$$\Re\left\{\boldsymbol{H}^\dagger\boldsymbol{H}\right\} = \begin{bmatrix} E_{xx} & E_{xy} \\ E_{xy} & E_{yy} \end{bmatrix} \tag{95}$$

where

$$E_{xx} = \Re\left\{\boldsymbol{h}_x^\dagger\boldsymbol{h}_x\right\} = \sum_{n=1}^{N}\sum_{k=1}^{K}\left|\frac{\partial y_n[k]}{\partial v_x}\right|^2 \tag{96}$$

$$E_{yy} = \Re\left\{\boldsymbol{h}_y^\dagger\boldsymbol{h}_y\right\} = \sum_{n=1}^{N}\sum_{k=1}^{K}\left|\frac{\partial y_n[k]}{\partial v_y}\right|^2 \tag{97}$$

$$E_{xy} = \Re\left\{\boldsymbol{h}_x^\dagger\boldsymbol{h}_y\right\} = \Re\left\{\sum_{n=1}^{N}\sum_{k=1}^{K}\left(\frac{\partial y_n[k]}{\partial v_x}\right)^*\frac{\partial y_n[k]}{\partial v_y}\right\}. \tag{98}$$

Here, E_{xx} and E_{yy} measure the sensitivity energy along the two horizontal directions, whereas E_{xy} measures the coupling between the two velocity components.

Similarly, the projected residual vector can be written as follows:

$$\Re\left\{\boldsymbol{H}^\dagger\Delta\boldsymbol{f}\right\} = \begin{bmatrix} P_x \\ P_y \end{bmatrix} \tag{99}$$

$$P_x = \Re\left\{\sum_{n=1}^{N}\sum_{k=1}^{K}\left(\frac{\partial y_n[k]}{\partial v_x}\right)^*\Delta f_n[k]\right\} \tag{100}$$

$$P_y = \Re\left\{\sum_{n=1}^{N}\sum_{k=1}^{K}\left(\frac{\partial y_n[k]}{\partial v_y}\right)^*\Delta f_n[k]\right\}. \tag{101}$$

Using the inversion formula for a 2×2 matrix gives the explicit perturbation estimates

$$\Delta\hat{v}_x = \frac{E_{yy}P_x - E_{xy}P_y}{E_{xx}E_{yy}-E_{xy}^2} \tag{102}$$

$$\Delta\hat{v}_y = \frac{E_{xx}P_y - E_{xy}P_x}{E_{xx}E_{yy}-E_{xy}^2}. \tag{103}$$

Hence, the locally linearized BLUE updates are

$$\hat{v}_{x,\text{BLUE}} = v_{x,\text{ref}} + \frac{E_{yy}P_x - E_{xy}P_y}{E_{xx}E_{yy}-E_{xy}^2} \tag{104}$$

$$\hat{v}_{y,\text{BLUE}} = v_{y,\text{ref}} + \frac{E_{xx}P_y - E_{xy}P_x}{E_{xx}E_{yy}-E_{xy}^2}. \tag{105}$$

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